



Population Health Management Report

Time Period:

Medical Data:2022-01-01 To 2024-04-01

Pharmacy Data:2022-07-01 To 2024-03-09

Prepared for

Long County

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1. Executive Summary

Introduction:

The following report is the result of an analysis of archival medical and pharmacy utilization data. The intent of this analysis is to yield a better understanding of the epidemiology currently influencing this population and to suggest population health management opportunities that can address the specific risk impacting this population. In order to accomplish this task, archival data was processed through proprietary algorithms in order to properly risk-stratify the population. The risk of a population has a direct relationship to current and future spending patterns. Variables that are the building blocks of risk and/or disease include, but are not limited to:

- Age, Gender, Lifestyle, Genetics, Chronic Illness, Co-Morbidities, Multi-Morbidities, Medication Compliance/Non-Compliance, Compliance/Non-Compliance to Evidence-Based Guidelines, Gaps in Care, etc.

The majority of the aforementioned variables were utilized to investigate risk stratifications within the population. The overall health of a population is determined by multiple factors; however, an individual's lifestyle is a powerful predictor of leading causes of morbidity and disability.

This analysis explored multiple areas of interest within the data, including the following research questions:

1. What is the cost burden of lifestyle modifiable risk factors within the employee population?
2. What is the relationship of age and gender to various disease states?
3. What are the gaps in care associated with suggested preventive measures for this population?
4. What is the relationship between drug compliance and non-compliance, as related to disease severity?
5. What is the financial burden associated with chronic disease within this population?
6. What is the level of HEDIS compliance (i.e., evidence-based & preventive medicine) within this population?
7. What is the expense related to specific co-morbidities (i.e., hypertension, hyperlipidemia, depression, etc.) within this population?
8. What variables best predict and explain future high spenders within this population?
9. What are actionable solutions that can be implemented to mitigate existing and future health risks?

Key Findings and Solutions for Consideration:

The following key findings resulted from the analysis of archival health care data.

Key Finding 1: Chronic Disease

Key Finding: Patterns of risk generally occur within any given population. In order to better understand these patterns, the population was risk stratified into five distinct groups:

Group	Description
Disease Group-1	No chronic disease and less than \$1,500 medical expenditures per 12 months
Disease Group-2	No chronic disease and \$1,500 or more medical expenditures per 12 months
Disease Group-3	One Chronic Disease
Disease Group-4	Two Chronic Disease
Disease Group-5	Three Chronic Disease
Disease Group-6	Four Chronic Disease
Disease Group-7	Five or More Chronic Disease

Total amount paid, Mean amount paid & Total patients (N) within the population for Risk Groups & Years were as follows (shown in chart below). As you would see in the analysis below; regarding the economic differences between each group; it reveals that mean expenditures increased as an individual incrementally progressed from a lower risk group to higher risk groups; from Group 3 - 7.

***N is a statistical notation that identifies the number of people in a population. Throughout this report, it is used to indicate*

the number of individuals incorporated into each analysis.

Risk Group Summary File Year	2022			2023			2024		
Risk Group Summary Risk Group	Risk Group Summary Total \$	Risk Group Summary Mean \$	Risk Group Summary N	Risk Group Summary Total \$	Risk Group Summary Mean \$	Risk Group Summary N	Risk Group Summary Total \$	Risk Group Summary Mean \$	Risk Group Summary N
GROUP-1	\$19,959	\$407	49	\$30,245	\$432	70	\$15,696	\$285	55
GROUP-2	\$87,538	\$5,471	16	\$180,545	\$7,523	24	\$19,126	\$2,732	7
GROUP-3	\$92,810	\$4,641	20	\$173,946	\$5,436	32	\$24,524	\$1,168	21
GROUP-4	\$49,846	\$7,121	7	\$69,632	\$9,947	7	\$2,587	\$431	6
GROUP-5	\$18,503	\$4,626	4	\$56,263	\$8,038	7			
GROUP-7	\$148,702	\$148,702	1						

Based on the chronic diseases included in the aforementioned Disease Group Risk Stratification, below are the top 3 Chronic diseases across all population who have a chronic disease. It would be estimated that an additional 10 to 15 percent of the population have chronic illness and have not yet been diagnosed, due to gaps in care.

The top 3 most expensive chronic diseases across populations & years were as follows (shown in chart below):

Medical records Reporting Year	2022	2023	2024
Medical records CHRONIC CATEGORY	Medical records TOTAL \$	Medical records TOTAL \$	Medical records TOTAL \$
DIABETES	\$7,224	\$4,052	\$1,318
CHRONIC KIDNEY DISEASE	\$1,360	\$1,356	\$946
HYPERTENSION	\$2,397	\$4,770	\$926
CATARACT	\$197	\$6,725	\$618
HYPERLIPIDEMIA	\$604	\$900	\$474
DEPRESSION	\$190	\$371	\$442
HEART DISEASE	\$0	\$42,693	\$372
OBESITY	\$68	\$424	\$121
ASTHMA	\$401	\$172	\$114
ATRIAL FIBRILLATION AND FLUTTER			\$61
PRESSURE AND CHRONIC ULCERS	\$476	\$11,979	\$59
LIVER DISEASE,CIRRHOSIS, AND OTHER LIVER CONDITIONS (EXCEPT VIRAL HEPATITIS)	\$441	\$296	\$35
CHRONIC OBSTRUCTIVE PULMONARY DISEASE	\$13,247	\$1,927	\$14
PERIPHERAL VASCULAR DISEASE (PVD)		\$199	
CANCER, BREAST		\$89	
OSTEOPOROSIS WITH OR WITHOUT PATHOLOGICAL FRACTURE	\$0	\$0	
STROKE/TRANSIENT ISCHEMIC ATTACK	\$0		

The top 3 most frequent chronic diseases across populations were:

Medical records Reporting Year	2022	2023	2024
Medical records CHRONIC CATEGORY	Medical records N	Medical records N	Medical records N
DIABETES	11	12	8
HYPERTENSION	17	20	7
DEPRESSION	3	4	4
HYPERLIPIDEMIA	6	9	4
ASTHMA	3	3	2
CHRONIC KIDNEY DISEASE	2	2	2
OBESITY	1	6	2
ATRIAL FIBRILLATION AND FLUTTER			1
CATARACT	1	1	1
CHRONIC OBSTRUCTIVE PULMONARY DISEASE	3	3	1
HEART DISEASE	2	4	1
LIVER DISEASE,CIRRHOSIS, AND OTHER LIVER CONDITIONS (EXCEPT VIRAL HEPATITIS)	2	2	1
PRESSURE AND CHRONIC ULCERS	1	1	1
OSTEOPOROSIS WITH OR WITHOUT PATHOLOGICAL FRACTURE	1	1	
STROKE/TRANSIENT ISCHEMIC ATTACK	1		
CANCER, BREAST		1	
PERIPHERAL VASCULAR DISEASE (PVD)		1	

Recommended Solution: The impact of chronic disease, co-morbidities, and disease-specific complications magnifies the impact of an individual's mean and overall expenditures. This type of stratification (i.e., the aforementioned Disease Group Risk Stratification) clearly shows that a relatively similar group of individuals drives a large percentage of overall expenditures. A population health management strategy that targets the low-risk or emerging risk portion of the population would potentially yield the highest return on investment.

The majority of wellness program strategies often do not implement programs that are sensitive to the clinical side of population health management and just concentrate on lifestyle modification (e.g., exercise, nutrition, stress management, etc.). However, in order to be effective with the chronic population, clinical strategies must be a part of the overall population health management strategy. Further analyses were conducted to identify the importance of chronic disease as a predictor of future spending.

Key Finding 2: Diabetes Complications and Co-Morbidities

Key Finding: The top three Diabetes-specific complications are shown in the chart below in order of their Total Spend across years.

Medical records Reporting Year	2022	2023	2024	Total
Medical records DIABETES SPECIFIC COMPLICATIONS	Medical records TOTAL \$	Medical records TOTAL \$	Medical records TOTAL \$	Medical records TOTAL \$
Cardiovascular	\$1,723	\$3,157		\$4,880
Peripheral Vascular		\$199		\$199
Neuropathy	\$84	\$40		\$124

Diabetes-specific complications are associated with uncontrolled diabetes and sometimes with undiagnosed diabetes. For example, a diagnosis of Idiopathic Neuropathy means "of no known cause"; however, it is often associated with an

undiagnosed case of diabetes. Wellness programming that includes biometric screenings would identify individuals with undiagnosed diabetes.

An analysis was conducted to determine the number of individuals with diabetes who were compliant with evidence-based guidelines for diabetes. The analysis revealed that there were a large number of individuals with a diagnosis of diabetes who are non-compliant to evidence-based medications related to diabetes management. Systems are available that can mail specific clinical “to dos” to each member’s home and monitor on-going compliance to these directions; this strategy also impacts the spouse and dependent children.

Recommended Solution: Establish evidence-based medicine guidelines (i.e., HEDIS goals) for the population that relate to diabetes management:

- Hemoglobin A1c (HbA1c) testing
- Hemoglobin A1c control (<7.0%)
- Retinal eye exam performed
- Screening for neuropathy
- Blood Pressure control (<130/80 mm/Hg)
- Medical attention for nephropathy

Key Finding 3: Preventive Screenings

Key Finding:

Preventive screenings for breast cancer, cervical cancer, and colorectal cancer were well below HEDIS National Guidelines. NCQA reports the following national screening rates:

- Breast Cancer Screening: Commercial HMO screening rate at 73.7% and Commercial PPO at 71.6%
- Cervical Cancer Screening: Commercial HMO screening rate at 76.2% and Commercial PPO at 74.2%
- Colorectal Cancer Screening: Commercial HMO screening rate at 65.0% and Commercial PPO at 61.8%

Actual screening rates for the population across years were as follows:

EBR Measures Year	EBR Measures Breast Cancer Screening Rate	EBR Measures Cervical Cancer Screening Rate	EBR Measures Colon Cancer Screening Rate
2022	40.0%	23.5%	13.0%
2023	70.0%	33.3%	0.0%
2024	22.2%	12.8%	10.0%

Recommended Solution: Increase the awareness of age/gender-specific preventive screenings within the population. Education in combination with various incentives would increase the population’s compliance with preventive screenings. Increased compliance to preventive screenings would identify diseases in the early stage, thus improving treatment outcomes and decreasing future expenditures.

Establish at least five HEDIS (Healthcare Effectiveness and Information Set) goals for the population. HEDIS is one of the most widely recognized healthcare performance measures in the United States. Suggested goals are as follows:

Goal 1: Increase the number of individuals between the ages of 18 to 75 who have a diagnosis of diabetes and are compliant with the following evidence-based medicine guidelines:

- Hemoglobin A1c (HbA1c) testing
- HbA1c poor control (>9.0%)
- HbA1c control (<8.0%)
- HbA1c control (<7.0%) for a selected population
- Eye exam (retinal) performed
- LDL-C screening
- LDL-C control (<100 mg/dl)

- Medical attention for nephropathy
- BP control (<130/80 mm Hg)

Goal 2: Increase the number of individuals between the ages of 18 to 74 who had an outpatient visit and had their body mass index (BMI) documented.

Goal 3: Increase the percentage of women between the ages of 40 to 69 who had a mammogram to screen for breast cancer.

Goal 4: Increase the percentage of women between the ages of 21 to 64 who received one or more Pap tests to screen for cervical cancer.

Goal 5: Increase the percentage of individuals between the ages of 50 to 75 who had an appropriate screening for colorectal cancer.

Key Finding 4: Musculoskeletal Diagnosis

Key Finding: It is our experience that; expenditures for Musculoskeletal-related diagnosis category are among the highest in any given population. The chart below shows the relative position of Musculoskeletal as a Disease category among Top 10 Diagnostic categories by Total Expenditure.

Medical records Reporting Year	2022		2023		2024	
Medical records DIAGNOSTIC CATEGORY	Medical records TOTAL \$	Medical records N	Medical records TOTAL \$	Medical records N	Medical records TOTAL \$	Medical records N
FACTORS INFLUENCING HEALTH STATUS AND CONTACT WITH HEALTH SERVICES	\$25,918	50	\$63,931	68	\$16,597	32
ENDOCRINE, NUTRITIONAL AND METABOLIC DISEASES	\$9,973	24	\$9,776	35	\$9,950	20
DISEASES OF THE DIGESTIVE SYSTEM	\$12,097	14	\$65,470	17	\$6,440	6
DISEASES OF THE MUSCULOSKELETAL SYSTEM AND CONNECTIVE TISSUE	\$71,603	29	\$22,274	35	\$5,460	11
DISEASES OF THE GENITOURINARY SYSTEM	\$2,958	7	\$57,596	22	\$5,214	12
SYMPTOMS, SIGNS AND ABNORMAL CLINICAL AND LABORATORY FINDINGS, NOT ELSEWHERE CLASSIFIED	\$53,522	57	\$45,394	64	\$4,344	20
DISEASES OF THE RESPIRATORY SYSTEM	\$19,693	27	\$35,970	39	\$2,197	11
MENTAL, BEHAVIORAL AND NEURODEVELOPMENTAL DISORDERS	\$5,468	20	\$8,889	27	\$1,569	16
DISEASES OF THE CIRCULATORY SYSTEM	\$25,173	21	\$51,449	25	\$1,439	10
CONGENITAL MALFORMATIONS, DEFORMATIONS AND CHROMOSOMAL ABNORMALITIES	\$3,758	4	\$2,650	5	\$1,246	2

A further analysis was completed to investigate which Musculoskeletal & Connective Tissue claims could potentially be work-related. Work-related musculoskeletal claims are usually associated with jobs or crafts that require manual material handling, frequent bending and twisting, static work posture, or whole body vibration. The results of this analysis were as follows (shown in chart below):

Medical records Reporting Year	2022	2023	2024
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Medical records WORK RELATED MSK DIAGNOSIS	Medical records TOTAL \$	Medical records MEAN \$	Medical records N	Medical records TOTAL \$	Medical records MEAN \$	Medical records N	Medical records TOTAL \$	Medical records MEAN \$	Medical records N
HAND & WRIST				\$307	\$307	1			
BACK	\$232	\$116	2						
UPPER EXTREMITY	\$138	\$138	1	\$34	\$34	1			

Recommended Solution: Based on the high frequency and costs associated with musculoskeletal medical claims, consider the implementation of pre-employment physical ability testing that simulates the essential functions of a particular job or craft. Conduct a job task analysis identify the essential functions of high-risk jobs. EEOC has specific guidelines for the design and implementation of physical ability tests. A well-designed physical ability test can help prevent worksite injury.

Key Finding 5: Medication Compliance

Key Finding: Calculation of a Medication Possession Ratio (MPR) revealed the following results for Hypertension medication and Statin (i.e., Lipid management) medication across population & years (shown in the chart below):

**N includes only those individuals who received a prescription and refilled it at least once.*

Medication Possession Ratio Year	Medication Possession Ratio Medication Possession Ratio	Medication Possession Ratio Total Members
2022	94.5%	56
2023	92.1%	99
2024	98.9%	67
Total	94.2%	117

Medication Possession Ratio Year	Medication Possession Ratio Medication Possession Ratio	Medication Possession Ratio Total Members
2022	87.8%	13
2023	89.7%	17
2024	98.2%	12
Total	91.1%	18

Medication Possession Ratio Year	Medication Possession Ratio Medication Possession Ratio	Medication Possession Ratio Total Members
2022	92.9%	13
2023	89.2%	26
2024	98.8%	19
Total	93.0%	33

The Medication Possession Ratio determines an individual's compliance to medications. However, it takes into account only those individuals who were prescribed medication and refilled the prescription at least once. It does not take into account the other people who may have a diagnosis, but no prescription has been tracked. For example, a person may have a diagnosis for hypertension, but they may not appear in the pharmacy data due to the fact that they either have no prescription or they have failed to fill a prescription they received.

Recommended Solution: Implement a solution that identifies all individuals who are non-compliant with medications and implement a mail-out reminder to the member's home address. Combine this strategy with an incentive connected to the member's benefit plan design.

Consider utilizing a therapeutic equivalent drug overlay to reduce pharmacy expenditures. A results-based wellness program will increase medication compliance. It is suggested that the current drug formulary within the health plan give incentive/disincentive toward value-based medication alternatives and therapeutic equivalent medications. Based on past analyses of other results-based wellness programs, many front-line drugs will be unnecessarily prescribed, thus driving expense.

Key Finding 6: Patient/Physician Communication

Key Finding: It should be noted as shown in the chart below that high frequencies of Symptoms, Signs, and ill-Defined Conditions could be a strong predictor of poor patient/physician communication. Within this category, no specific diagnosis is rendered, yet treatment cost is experienced. For example, with a diagnosis of Symptoms, Signs, and Ill-Defined Conditions involving the abdomen, in reality the diagnosis could be more specific as Gastro Esophageal Reflux Disease (GERD).

Recommended Solution: Personal electronic health records can help improve the accuracy of an individual’s diagnosis, and writing down all symptoms prior to a physician visit can also improve the accuracy of diagnosis.

Total Spend (\$) & Patients (N) - SYMPTOMS, SIGNS, AND ABNORMAL CLINICAL LABORATORY FINDINGS

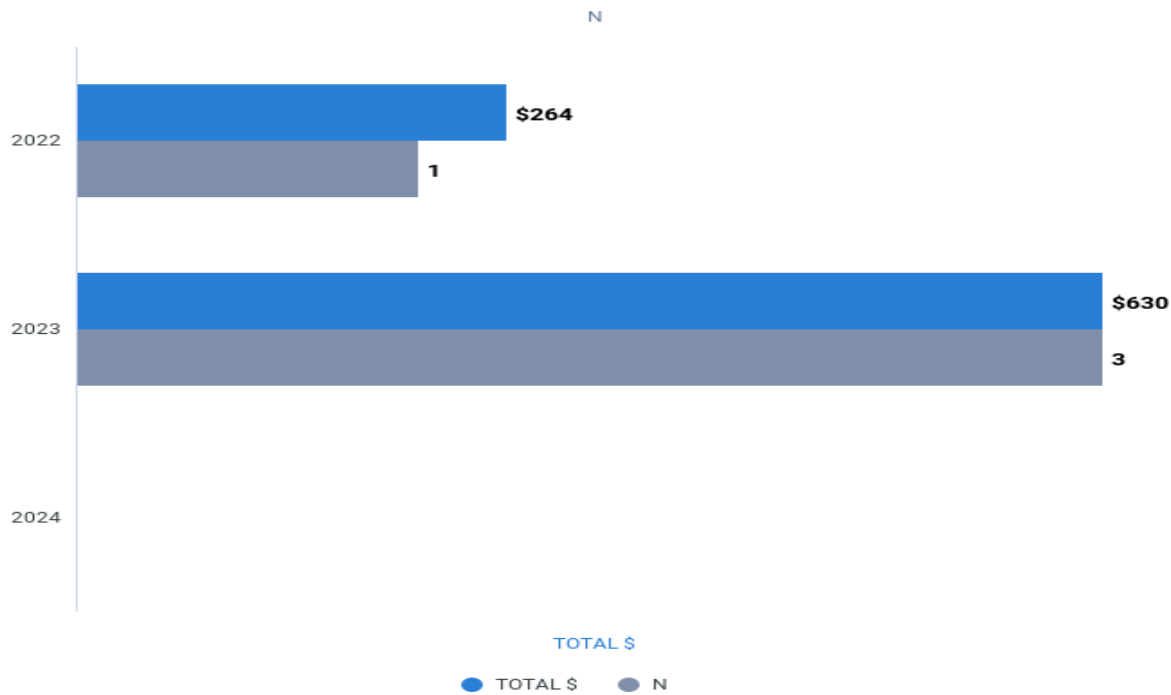
Medical records Reporting Year	Medical records TOTAL \$	Medical records MEAN \$	Medical records N
2022	\$53,522	\$939	57
2023	\$45,394	\$709	64
2024	\$4,344	\$217	20

Key Finding 7: Avoidable Emergency Room Visits

Key Finding: Avoidable ER visits are defined as those visits which could have been appropriately treated in another setting at the time the visit occurred. The State of Washington, through sampling of 53 hospitals and 2.2 million patients, established the definition of avoidable ER visits. Avoidable ER visits have the following statistics:

- 1 out of 9 visits is avoidable.
- Avoidable visits account for approximately 11 percent of the overall ER spend.
- Children that are less than 18 years of age comprise 1/3 of all avoidable visits.
- The majority of avoidable visits are comprised of females.
- The uninsured have approximately the same rate of avoidable visits as compared to the insured.
- The majority of avoidable ER visits occur between 12 p.m. and 8 p.m.

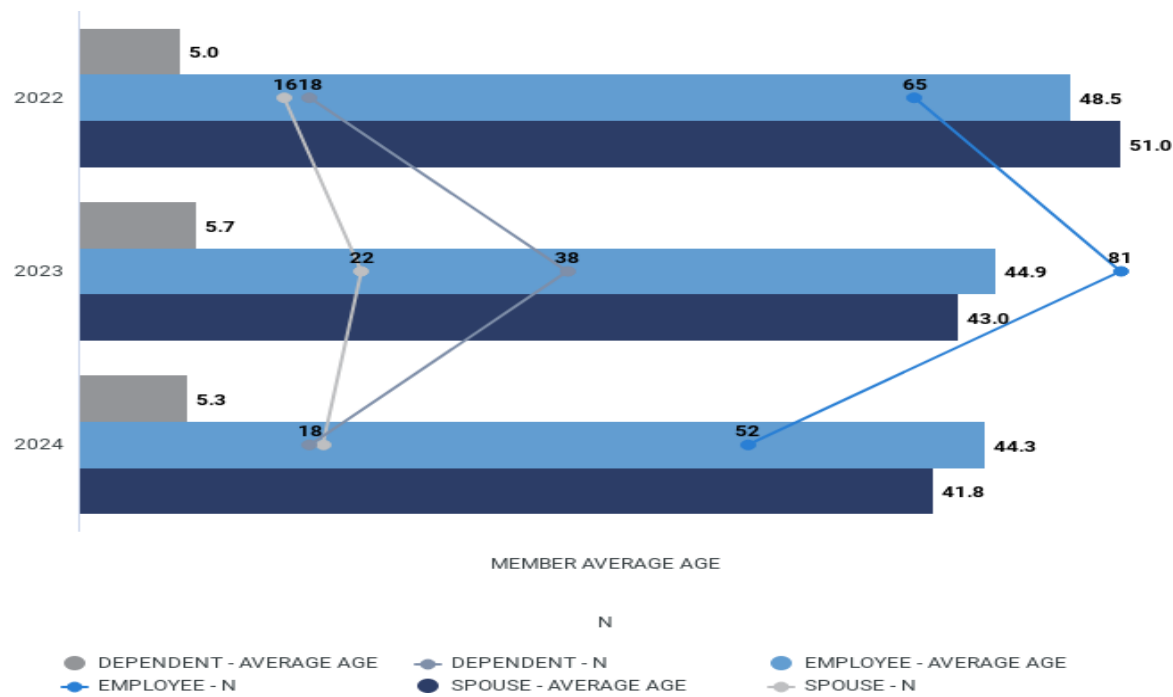
Total spend for Avoidable ER disease across population & years were as follows (shown in chart below):



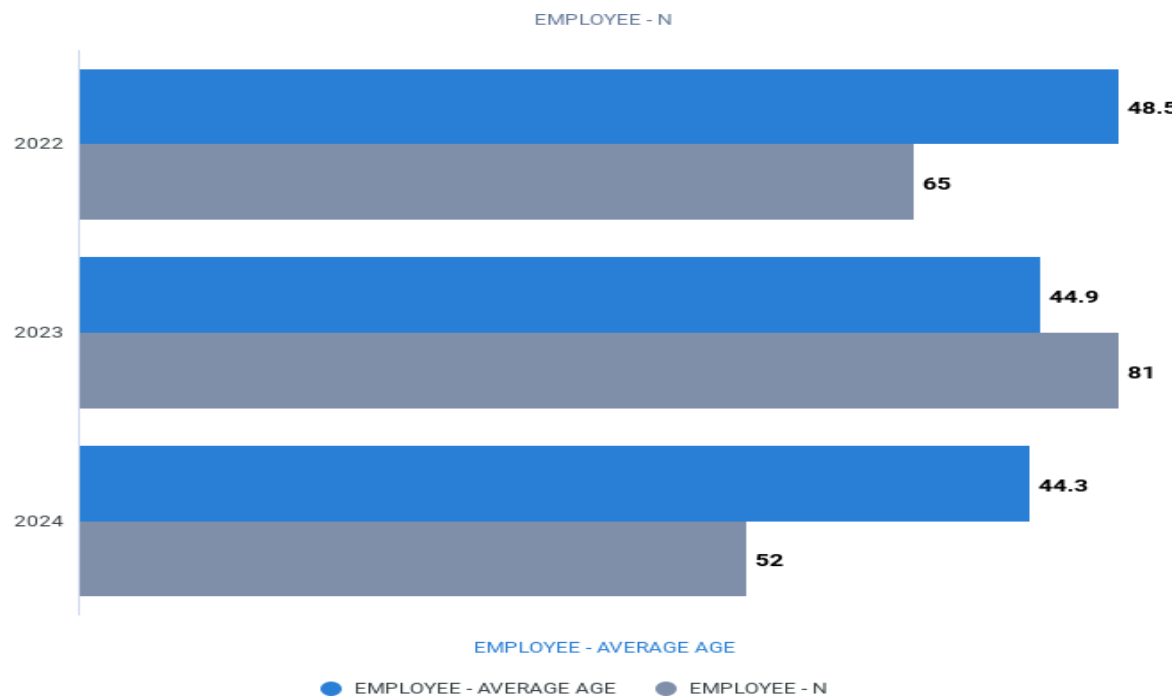
Recommended Solution: In order to effectively reduce avoidable ER visits, frequent flyers need to be identified and connected with a primary care physician. The State of Washington research indicated that if these individuals are assigned a primary care physician, avoidable ER visits will be reduced by approximately 58 percent. It would also be suggested to distribute medical self-care guides to help people differentiate between an emergency and a situation that can be resolved at an alternative setting. One other leading cause for avoidable ER visits is related to drug seeking behavior; this can be limited by urging hospitals to limit the amount of pain management drugs that are prescribed, especially opioid-based medications.

2. Demographic Information (Age and Gender)

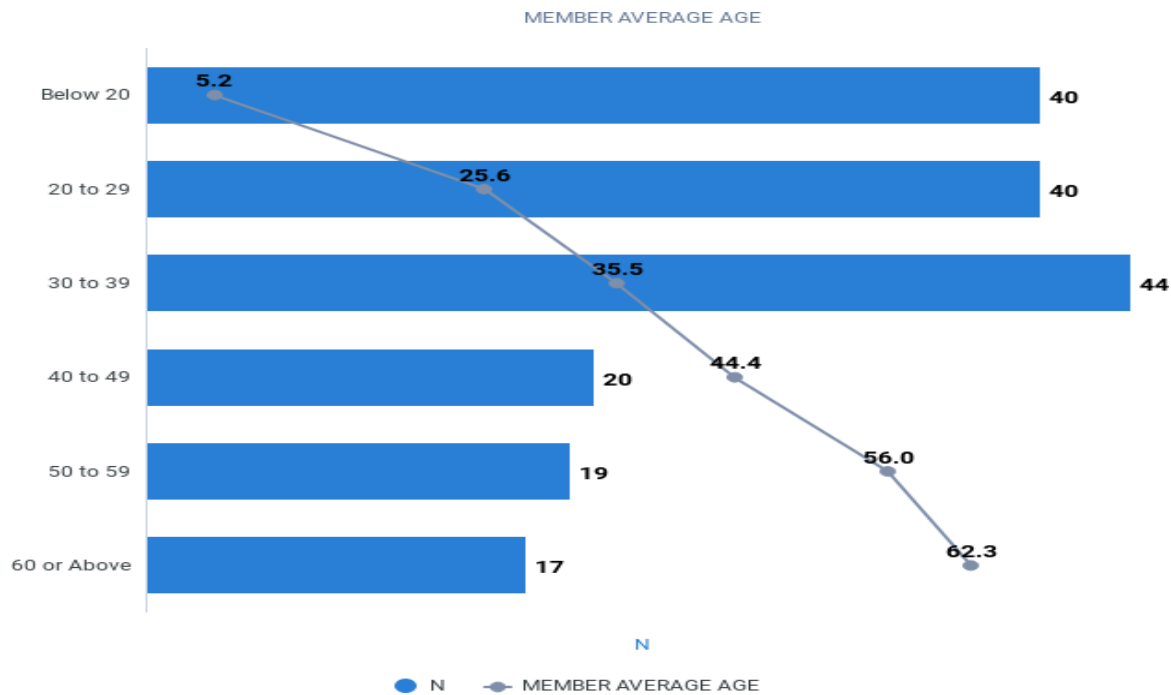
- The mean (average) age for the total population (including Employees, Spouses, and Dependents) was:



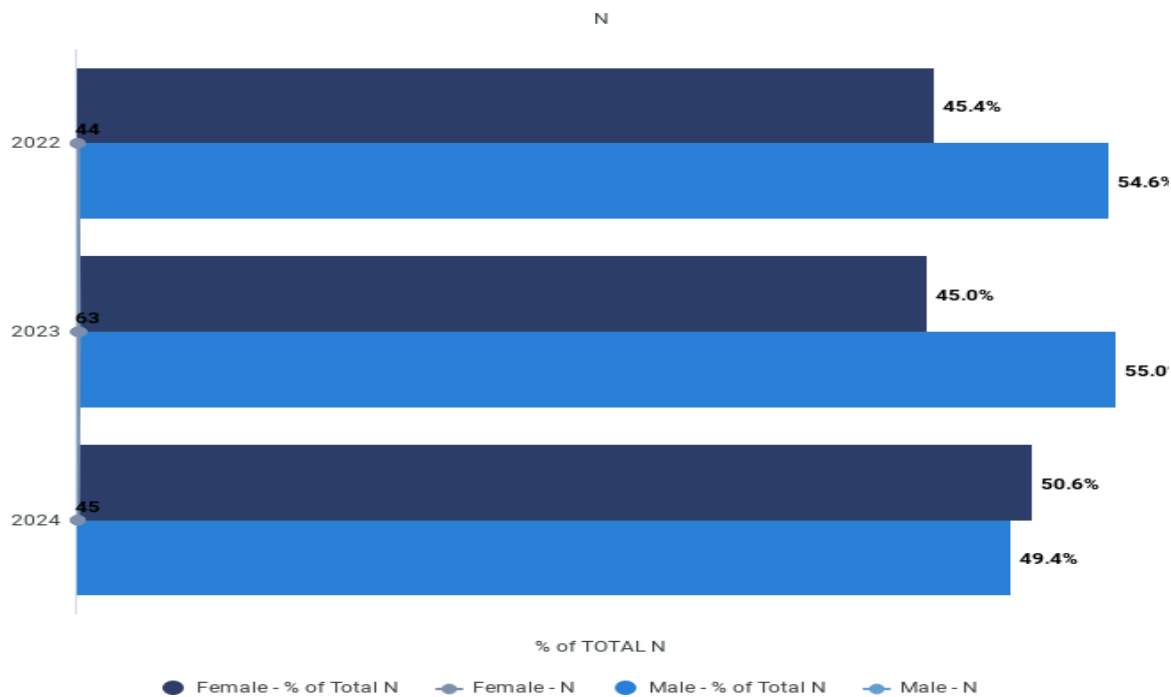
- For employees only, the mean age was:



- Number of Individuals by Age - Total Population (Employee, Spouse & Dependent):



- The gender breakdown for the total population (Employee, Spouse & Dependent) was:



3. Overall Medical Expenditures by Year

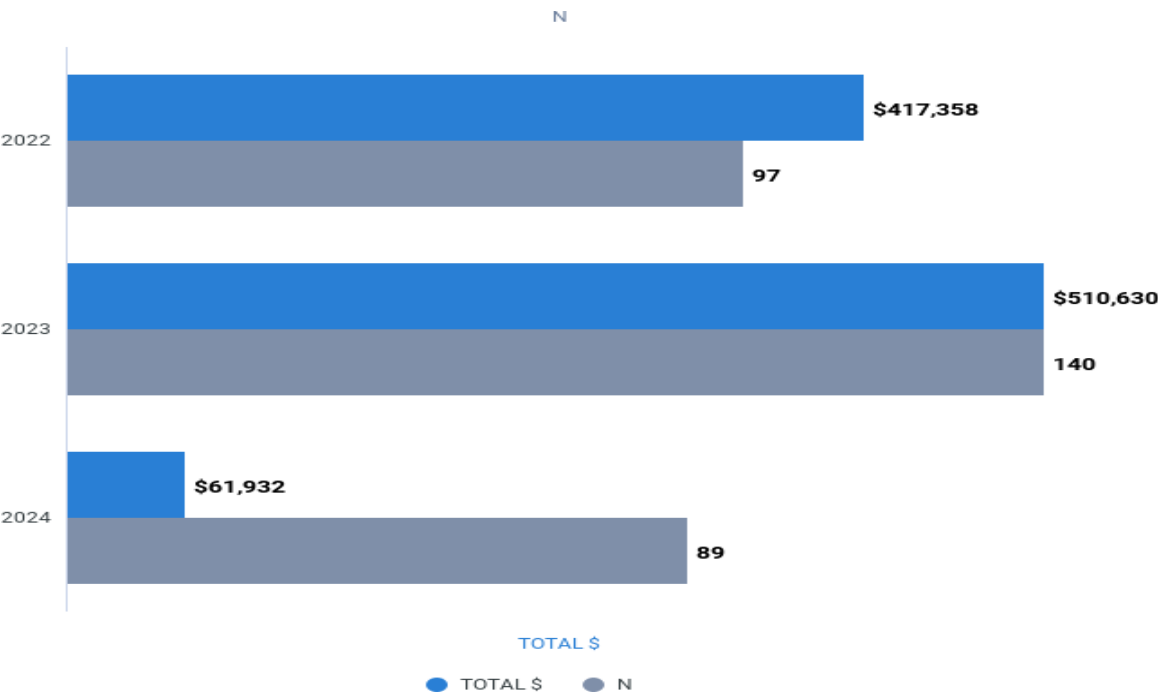
Medical expenditures* (based on paid claims) were as follows:

Medical records Reporting Year	Medical records TOTAL \$	Medical records MEAN \$	Medical records N
2022	\$417,358	\$4,303	97
2023	\$510,630	\$3,647	140
2024	\$61,932	\$696	89
Total	\$989,921	\$6,000	165

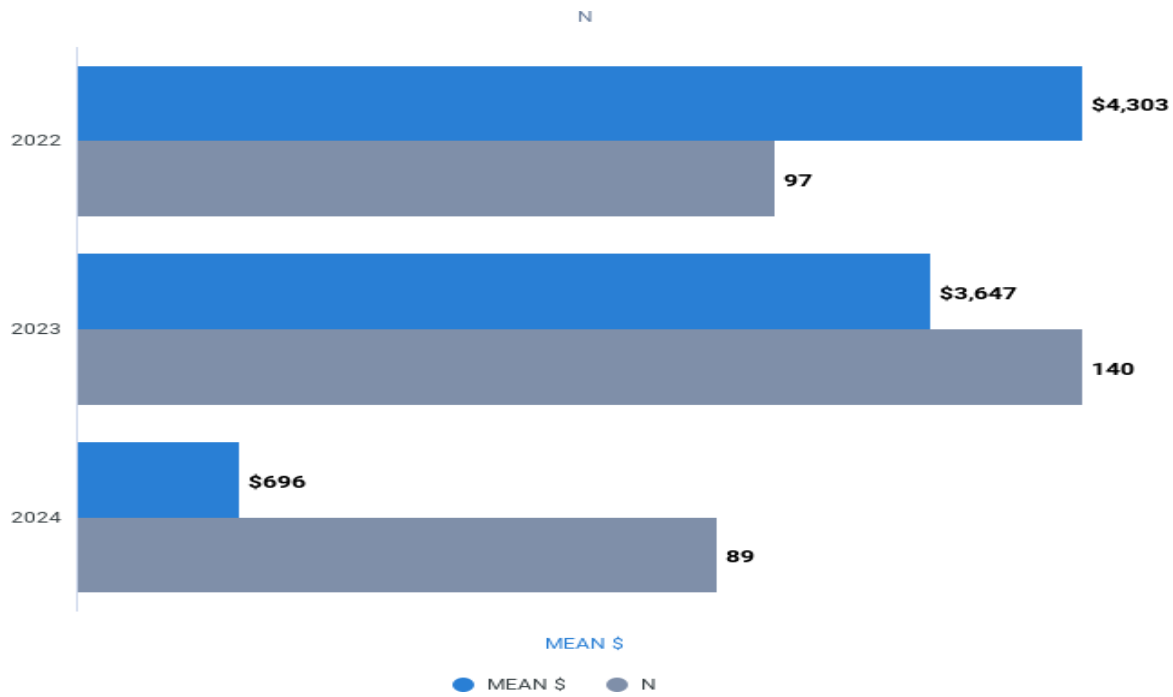
*Medical expenditures do not include pharmacy-related expenditures.

**N is a statistical notation that identifies the number of people in a population. Throughout this report, it is used to indicate the number of individuals incorporated into each analysis.

Total Amount Paid - Medical - Total Population

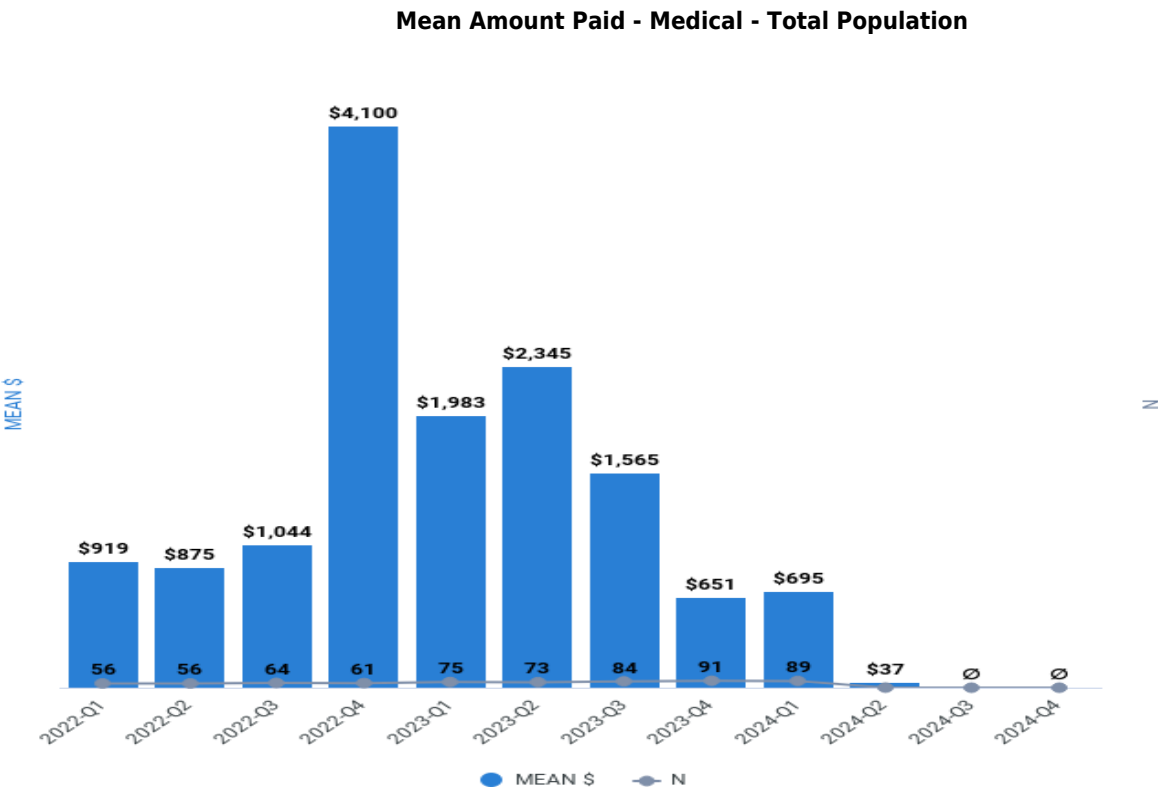
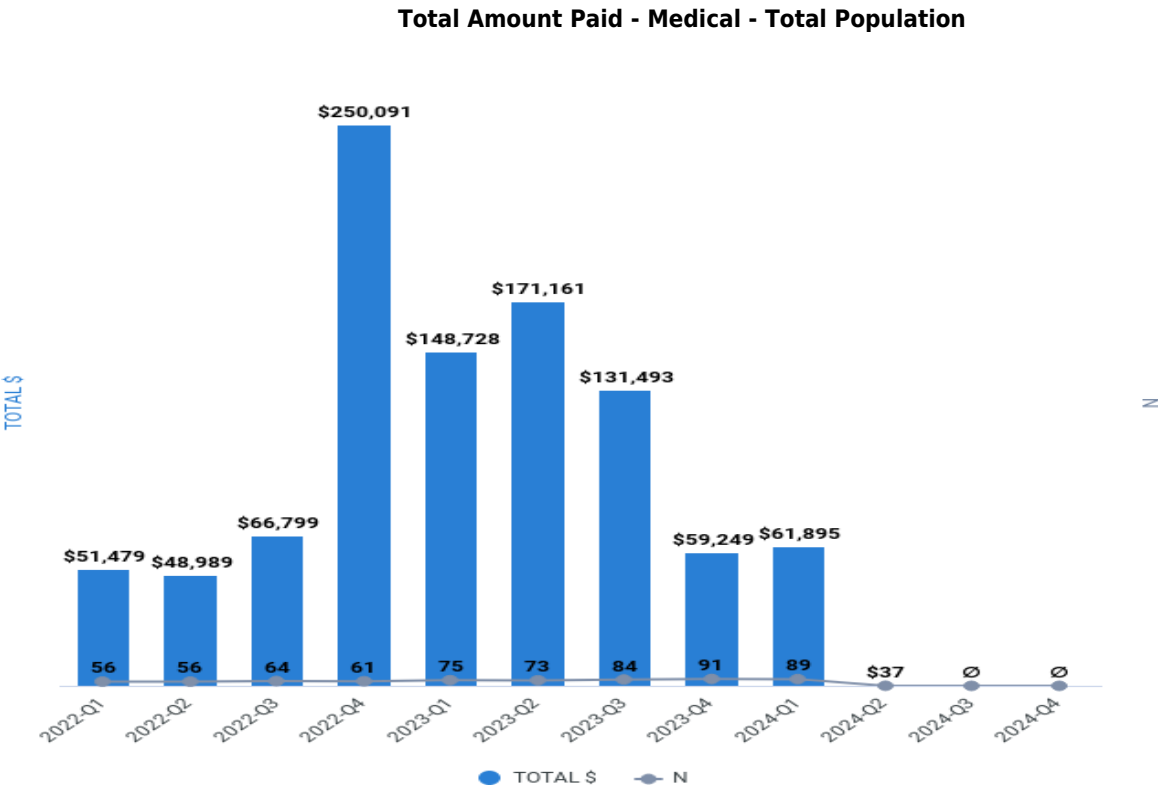


Mean Amount Paid - Medical - Total Population



4. Overall Medical Expenditures by Quarter

Medical expenditures were as follows:

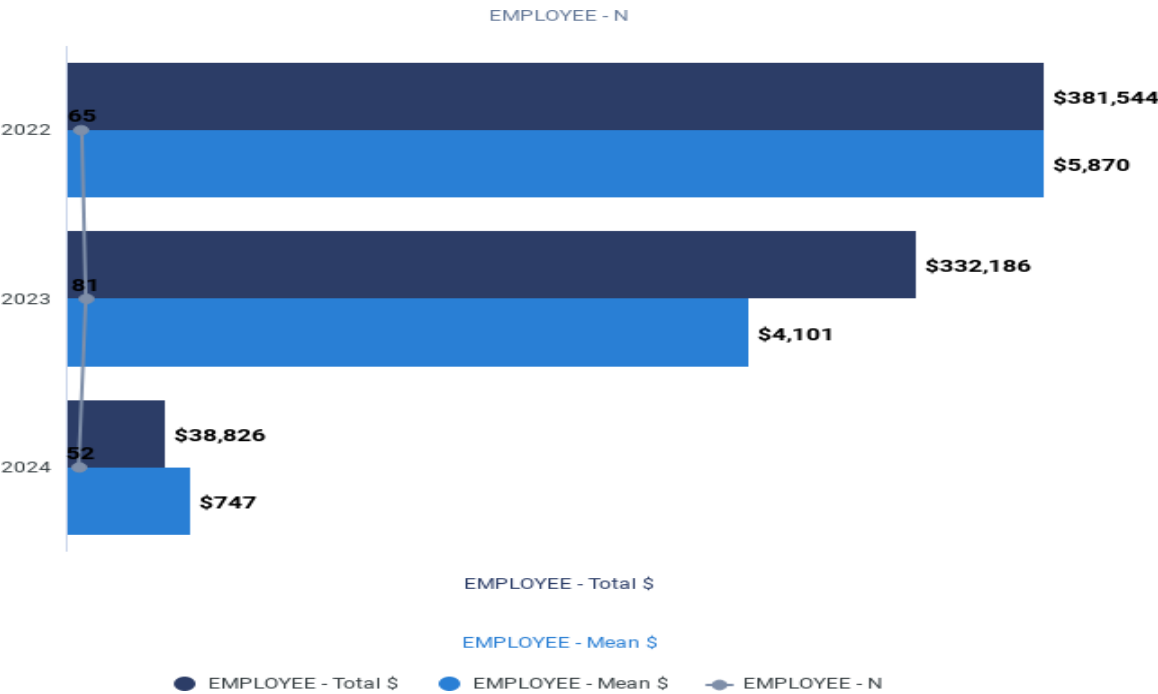


5. Employee/ Spouse/ Dependent Expenditures

Medical expenditures related to Employees, Spouses, and Dependents were as follows:

Medical records Reporting Year	2022			2023			2024		
Medical records RELATIONSHIP TO EMPLOYEE	Medical records TOTAL \$	Medical records MEAN \$	Medical records N	Medical records TOTAL \$	Medical records MEAN \$	Medical records N	Medical records TOTAL \$	Medical records MEAN \$	Medical records N
EMPLOYEE	\$381,544	\$5,870	65	\$332,186	\$4,101	81	\$38,826	\$747	52
SPOUSE	\$14,756	\$922	16	\$115,978	\$5,272	22	\$14,941	\$786	19
DEPENDENT	\$21,058	\$1,170	18	\$62,467	\$1,644	38	\$8,165	\$454	18

Total Amount Paid - Medical - Employee / Spouse / Dependent

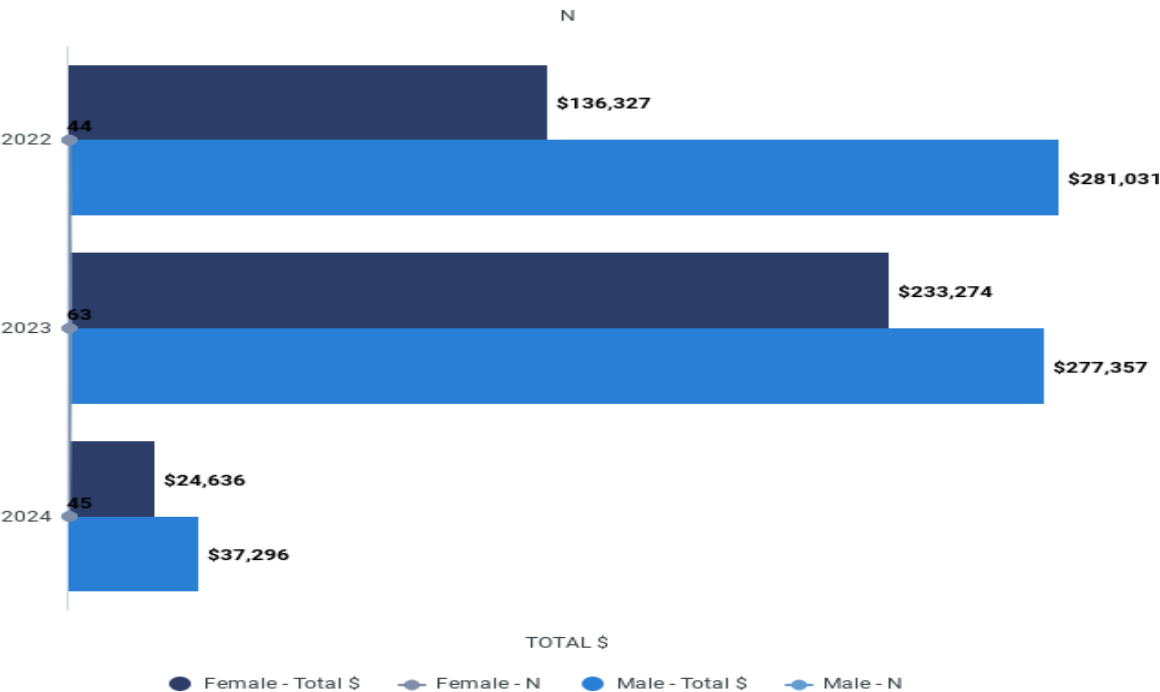


6. Gender Related Expenditures

Medical expenditures related to Males and Females were as follows:

Medical records MEMBER GENDER	F			M		
Medical records Reporting Year	Medical records TOTAL \$	Medical records MEAN \$	Medical records N	Medical records TOTAL \$	Medical records MEAN \$	Medical records N
2022	\$136,327	\$3,098	44	\$281,031	\$5,302	53
2023	\$233,274	\$3,703	63	\$277,357	\$3,602	77
2024	\$24,636	\$547	45	\$37,296	\$848	44

Total Amount Paid - Gender - Total Population



7. Diagnostic Category Expenditures

Diagnostic categories* ranked by expense were as follows:

*Refer to Appendix 2 for examples of Diagnostic Categories.

Medical records Reporting Year	2022		2023		2024	
Medical records DIAGNOSTIC CATEGORY	Medical records TOTAL \$	Medical records N	Medical records TOTAL \$	Medical records N	Medical records TOTAL \$	Medical records N
FACTORS INFLUENCING HEALTH STATUS AND CONTACT WITH HEALTH SERVICES	\$25,918	50	\$63,931	68	\$16,597	32
ENDOCRINE, NUTRITIONAL AND METABOLIC DISEASES	\$9,973	24	\$9,776	35	\$9,950	20
DISEASES OF THE DIGESTIVE SYSTEM	\$12,097	14	\$65,470	17	\$6,440	6
DISEASES OF THE MUSCULOSKELETAL SYSTEM AND CONNECTIVE TISSUE	\$71,603	29	\$22,274	35	\$5,460	11
DISEASES OF THE GENITOURINARY SYSTEM	\$2,958	7	\$57,596	22	\$5,214	12
SYMPTOMS, SIGNS AND ABNORMAL CLINICAL AND LABORATORY FINDINGS, NOT ELSEWHERE CLASSIFIED	\$53,522	57	\$45,394	64	\$4,344	20
DISEASES OF THE RESPIRATORY SYSTEM	\$19,693	27	\$35,970	39	\$2,197	11
MENTAL, BEHAVIORAL AND NEURODEVELOPMENTAL DISORDERS	\$5,468	20	\$8,889	27	\$1,569	16
DISEASES OF THE CIRCULATORY SYSTEM	\$25,173	21	\$51,449	25	\$1,439	10
CONGENITAL MALFORMATIONS, DEFORMATIONS AND CHROMOSOMAL ABNORMALITIES	\$3,758	4	\$2,650	5	\$1,246	2
DISEASES OF THE SKIN AND SUBCUTANEOUS TISSUE	\$5,345	15	\$15,447	13	\$1,209	7
INJURY, POISONING, AND CERTAIN OTHER CONSEQUENCES OF EXTERNAL CAUSES	\$15,500	12	\$26,528	11	\$1,065	4
DISEASES OF THE NERVOUS SYSTEM	\$828	10	\$5,294	13	\$926	9
DISEASES OF THE EYE AND ADNEXA	\$768	4	\$9,276	12	\$907	3
PREGNANCY, CHILDBIRTH AND THE PUERPERIUM	\$0	1	\$55,583	4	\$845	2
NEOPLASMS	\$143,043	10	\$8,532	12	\$706	2
DISEASES OF THE EAR AND MASTOID PROCESS	\$7,680	10	\$1,897	11	\$470	4
CODES FOR SPECIAL PURPOSES	\$6,995	11	\$3,364	4	\$416	3
CERTAIN INFECTIONS AND PARASITIC DISEASES	\$4,115	6	\$8,071	17	\$178	2
DISEASES OF THE BLOOD AND BLOOD-FORMING ORGANS AND CERTAIN DISORDERS INVOLVING THE IMMUNE MECHANISM	\$1,327	3	\$513	5	\$104	2
CERTAIN CONDITIONS ORIGINATING IN THE PERINATAL PERIOD			\$2,313	1		

Mean Amount Paid - Diagnostic Category - Total Population

Medical records Reporting Year	2022		2023		2024	
Medical records DIAGNOSTIC CATEGORY	Medical records MEAN \$	Medical records N	Medical records MEAN \$	Medical records N	Medical records MEAN \$	Medical records N
DISEASES OF THE DIGESTIVE SYSTEM	\$864	14	\$3,851	17	\$1,073	6
CONGENITAL MALFORMATIONS, DEFORMATIONS AND CHROMOSOMAL ABNORMALITIES	\$939	4	\$530	5	\$623	2
FACTORS INFLUENCING HEALTH STATUS AND CONTACT WITH HEALTH SERVICES	\$518	50	\$940	68	\$519	32
ENDOCRINE, NUTRITIONAL AND METABOLIC DISEASES	\$416	24	\$279	35	\$497	20
DISEASES OF THE MUSCULOSKELETAL SYSTEM AND CONNECTIVE TISSUE	\$2,469	29	\$636	35	\$496	11
DISEASES OF THE GENITOURINARY SYSTEM	\$423	7	\$2,618	22	\$434	12
PREGNANCY, CHILDBIRTH AND THE PUERPERIUM	\$0	1	\$13,896	4	\$422	2
NEOPLASMS	\$14,304	10	\$711	12	\$353	2
DISEASES OF THE EYE AND ADNEXA	\$192	4	\$773	12	\$302	3
INJURY, POISONING, AND CERTAIN OTHER CONSEQUENCES OF EXTERNAL CAUSES	\$1,292	12	\$2,412	11	\$266	4
SYMPTOMS, SIGNS AND ABNORMAL CLINICAL AND LABORATORY FINDINGS, NOT ELSEWHERE CLASSIFIED	\$939	57	\$709	64	\$217	20
DISEASES OF THE RESPIRATORY SYSTEM	\$729	27	\$922	39	\$200	11
DISEASES OF THE SKIN AND SUBCUTANEOUS TISSUE	\$356	15	\$1,188	13	\$173	7
DISEASES OF THE CIRCULATORY SYSTEM	\$1,199	21	\$2,058	25	\$144	10
CODES FOR SPECIAL PURPOSES	\$636	11	\$841	4	\$139	3
DISEASES OF THE EAR AND MASTOID PROCESS	\$768	10	\$172	11	\$117	4
DISEASES OF THE NERVOUS SYSTEM	\$83	10	\$407	13	\$103	9
MENTAL, BEHAVIORAL AND NEURODEVELOPMENTAL DISORDERS	\$273	20	\$329	27	\$98	16
CERTAIN INFECTIONS AND PARASITIC DISEASES	\$686	6	\$475	17	\$89	2
DISEASES OF THE BLOOD AND BLOOD-FORMING ORGANS AND CERTAIN DISORDERS INVOLVING THE IMMUNE MECHANISM	\$442	3	\$103	5	\$52	2
CERTAIN CONDITIONS ORIGINATING IN THE PERINATAL PERIOD			\$2,313	1		

8. Chronic Disease Expenditures

Chronic diseases* ranked by expense were as follows:

**For this calculation, if an individual has multiple chronic diseases, they will be counted for each chronic disease.*

Medical records Reporting Year	2022		2023		2024	
Medical records CHRONIC CATEGORY	Medical records TOTAL \$	Medical records N	Medical records TOTAL \$	Medical records N	Medical records TOTAL \$	Medical records N
DIABETES	\$7,224	11	\$4,052	12	\$1,318	8
CHRONIC KIDNEY DISEASE	\$1,360	2	\$1,356	2	\$946	2
HYPERTENSION	\$2,397	17	\$4,770	20	\$926	7
CATARACT	\$197	1	\$6,725	1	\$618	1
HYPERLIPIDEMIA	\$604	6	\$900	9	\$474	4
DEPRESSION	\$190	3	\$371	4	\$442	4
HEART DISEASE	\$0	2	\$42,693	4	\$372	1
OBESITY	\$68	1	\$424	6	\$121	2
ASTHMA	\$401	3	\$172	3	\$114	2
ATRIAL FIBRILLATION AND FLUTTER					\$61	1
PRESSURE AND CHRONIC ULCERS	\$476	1	\$11,979	1	\$59	1
LIVER DISEASE,CIRRHOSIS, AND OTHER LIVER CONDITIONS (EXCEPT VIRAL HEPATITIS)	\$441	2	\$296	2	\$35	1
CHRONIC OBSTRUCTIVE PULMONARY DISEASE	\$13,247	3	\$1,927	3	\$14	1
PERIPHERAL VASCULAR DISEASE (PVD)			\$199	1		
CANCER, BREAST			\$89	1		
OSTEOPOROSIS WITH OR WITHOUT PATHOLOGICAL FRACTURE	\$0	1	\$0	1		
STROKE/TRANSIENT ISCHEMIC ATTACK	\$0	1				

Mean Amount Paid - Chronic Category - Total Population

Medical records Reporting Year	2022		2023		2024	
Medical records CHRONIC CATEGORY	Medical records MEAN \$	Medical records N	Medical records MEAN \$	Medical records N	Medical records MEAN \$	Medical records N
CATARACT	\$197	1	\$6,725	1	\$618	1
CHRONIC KIDNEY DISEASE	\$680	2	\$678	2	\$473	2
HEART DISEASE	\$0	2	\$10,673	4	\$372	1
DIABETES	\$657	11	\$338	12	\$165	8
HYPERTENSION	\$141	17	\$239	20	\$132	7
HYPERLIPIDEMIA	\$101	6	\$100	9	\$119	4
DEPRESSION	\$63	3	\$93	4	\$110	4
ATRIAL FIBRILLATION AND FLUTTER					\$61	1
OBESITY	\$68	1	\$71	6	\$60	2

PRESSURE AND CHRONIC ULCERS	\$476	1	\$11,979	1	\$59	1
ASTHMA	\$134	3	\$57	3	\$57	2
LIVER DISEASE,CIRRHOSIS, AND OTHER LIVER CONDITIONS (EXCEPT VIRAL HEPATITIS)	\$220	2	\$148	2	\$35	1
CHRONIC OBSTRUCTIVE PULMONARY DISEASE	\$4,416	3	\$642	3	\$14	1
PERIPHERAL VASCULAR DISEASE (PVD)			\$199	1		
CANCER, BREAST			\$89	1		
OSTEOPOROSIS WITH OR WITHOUT PATHOLOGICAL FRACTURE	\$0	1	\$0	1		
STROKE/TRANSIENT ISCHEMIC ATTACK	\$0	1				

Chronic diseases* ranked by frequency were as follows:

* For this calculation, if an individual has multiple chronic diseases, they will be counted for each chronic disease.

Medical records Reporting Year	2022	2023	2024
Medical records CHRONIC CATEGORY	Medical records N	Medical records N	Medical records N
DIABETES	11	12	8
HYPERTENSION	17	20	7
DEPRESSION	3	4	4
HYPERLIPIDEMIA	6	9	4
ASTHMA	3	3	2
CHRONIC KIDNEY DISEASE	2	2	2
OBESITY	1	6	2
ATRIAL FIBRILLATION AND FLUTTER			1
CATARACT	1	1	1
CHRONIC OBSTRUCTIVE PULMONARY DISEASE	3	3	1
HEART DISEASE	2	4	1
LIVER DISEASE,CIRRHOSIS, AND OTHER LIVER CONDITIONS (EXCEPT VIRAL HEPATITIS)	2	2	1
PRESSURE AND CHRONIC ULCERS	1	1	1
OSTEOPOROSIS WITH OR WITHOUT PATHOLOGICAL FRACTURE	1	1	
STROKE/TRANSIENT ISCHEMIC ATTACK	1		
CANCER, BREAST		1	
PERIPHERAL VASCULAR DISEASE (PVD)		1	

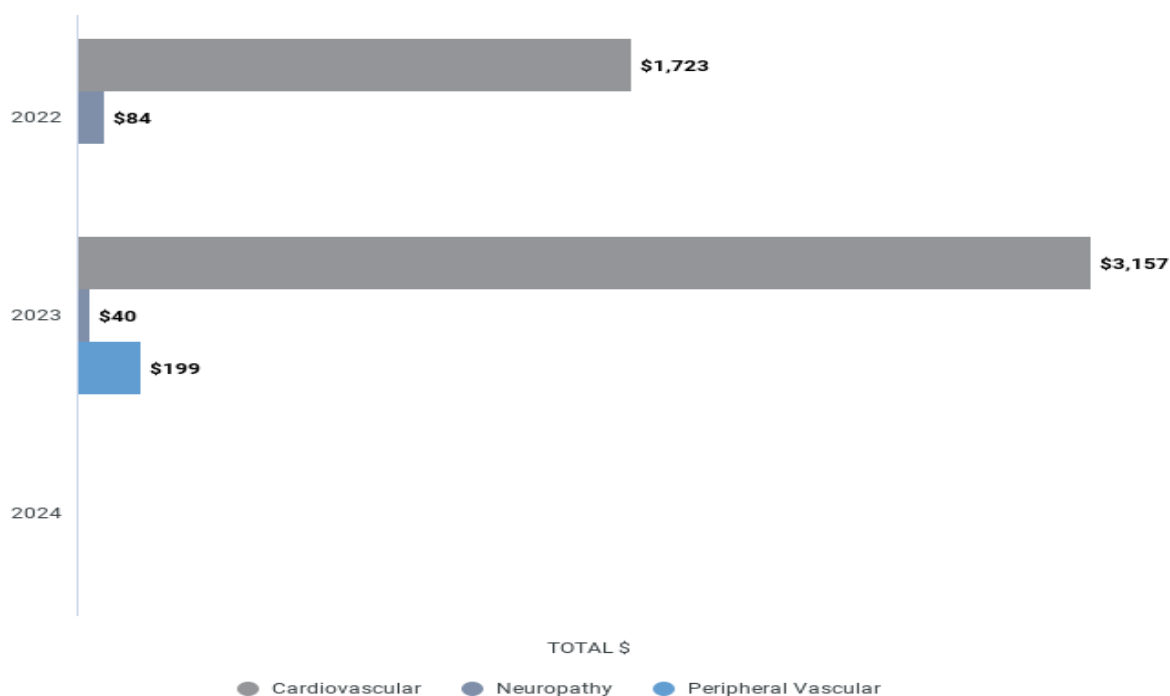
9. Diabetes Expenditures & Related Risk Stratification

Diabetes expensive complications* ranked by expense were as follows:

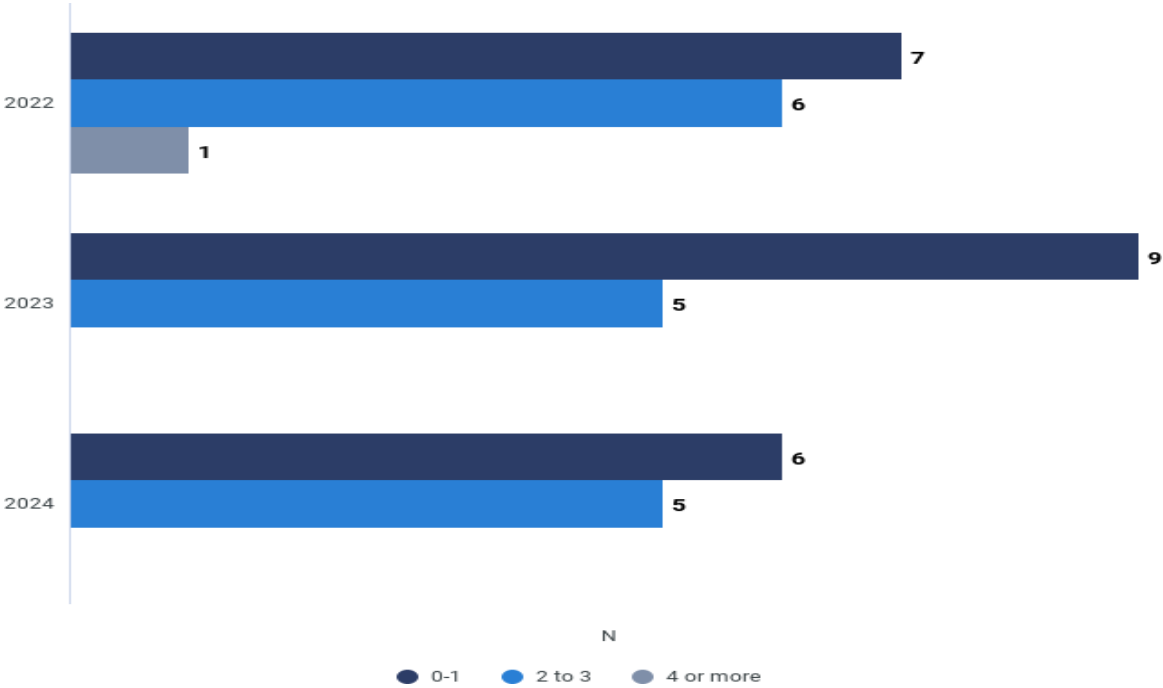
*Refer to Appendix 3 for examples of Complications of Diabetes.

Medical records Reporting Year	2022		2023		2024	
Medical records DIABETES SPECIFIC COMPLICATIONS	Medical records TOTAL \$	Medical records N	Medical records TOTAL \$	Medical records N	Medical records TOTAL \$	Medical records N
Cardiovascular	\$1,723	5	\$3,157	5		
Peripheral Vascular			\$199	1		
Neuropathy	\$84	1	\$40	1		

Total Amount Paid - Complications of Diabetes - Total Population



Number of Individuals with a Diagnosis of Diabetes in Combination with Co-Morbidities



10. Diabetes Non-Compliance to Evidence-Based Medicine

The following data identifies the number of individuals who had a diagnosis of diabetes and were non-compliant with evidence-based medicine guidelines.

**Eligibility is defined as having a diagnosis of Diabetes or diagnosis of Diabetes with End Organ Damage.*

Ad hoc Tool Service Year	Ad hoc Tool Diabetes, No Ace Inhibiter	Ad hoc Tool Diabetes, No ARB	Ad hoc Tool Diabetes, No DRI	Ad hoc Tool Diabetes, No Statin Drug
2022	7	10	11	3
2023	8	11	12	6
2024	5	5	8	5
Total	9	12	15	8

Medical records Reporting Year	Medical records Diabetes, No Foot Exam	Medical records Diabetes, No Eye Exam	Medical records Diabetes, No HbA1c
2022	11	11	11
2023	12	12	12
2024	8	7	8
Total	15	15	15

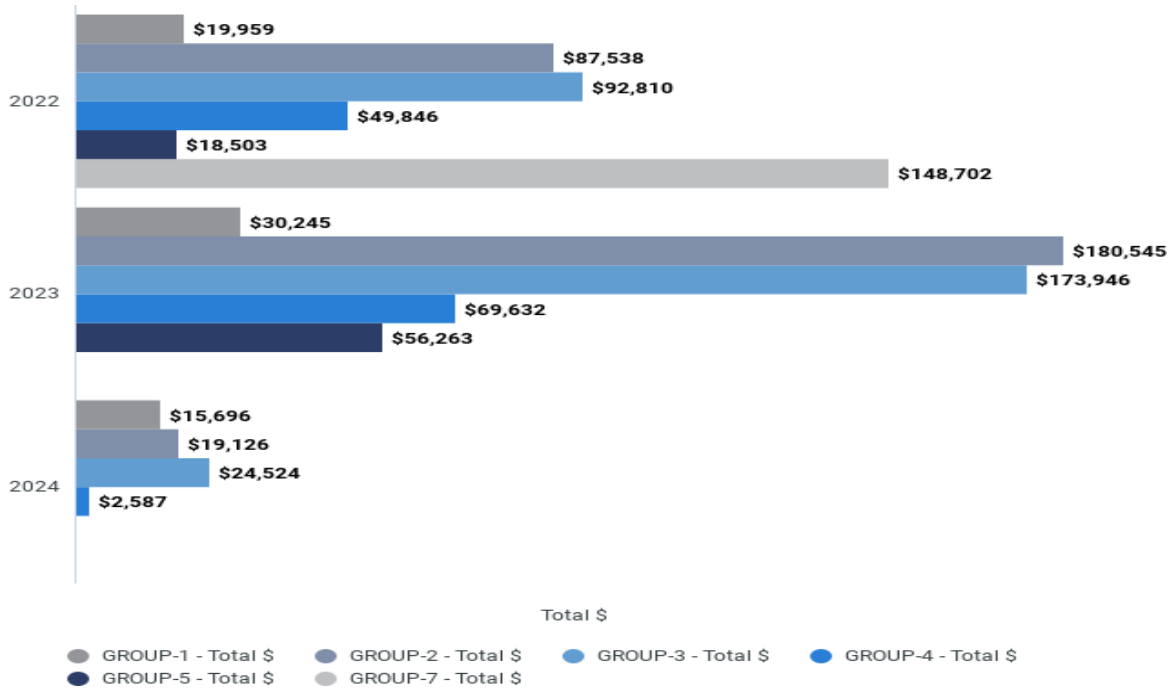
11. Disease Group Risk Stratification

The population was risk stratified into the following seven distinct groups:

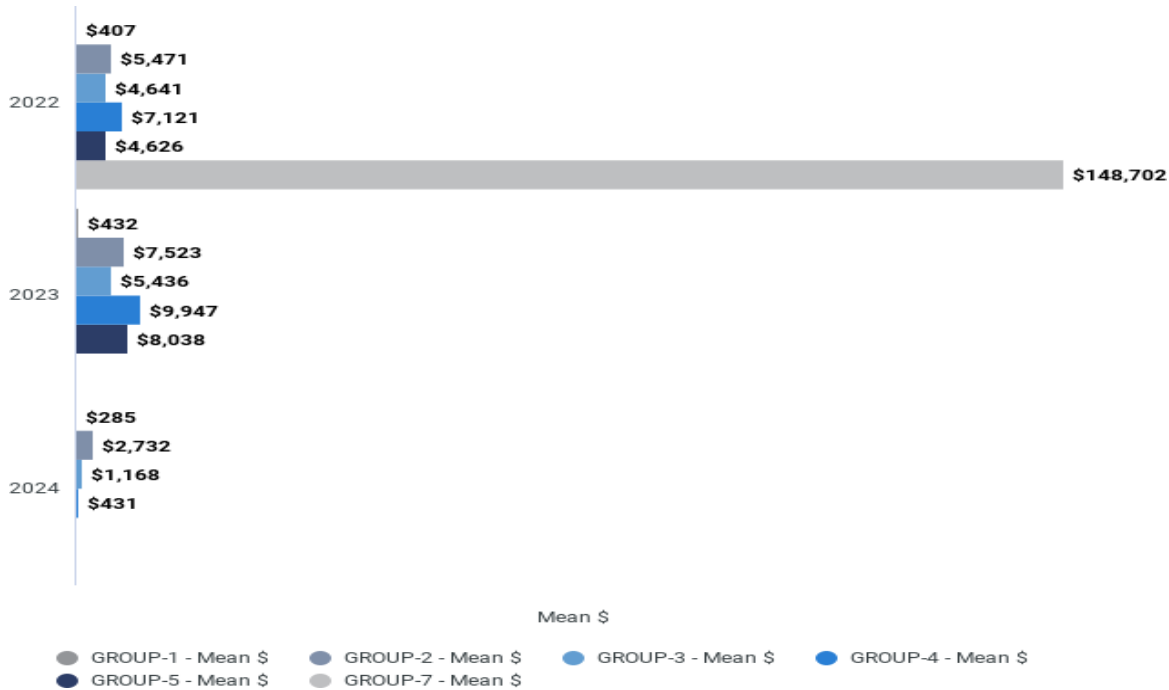
Group	Description
Disease Group-1	No chronic disease and less than \$1,500 medical expenditures per 12 months
Disease Group-2	No chronic disease and \$1,500 or more medical expenditures per 12 months
Disease Group-3	One Chronic Disease
Disease Group-4	Two Chronic Disease
Disease Group-5	Three Chronic Disease
Disease Group-6	Four Chronic Disease
Disease Group-7	Five or More Chronic Disease

Risk Group Summary File Year	2022			2023			2024		
Risk Group Summary Risk Group	Risk Group Summary Total \$	Risk Group Summary Mean \$	Risk Group Summary N	Risk Group Summary Total \$	Risk Group Summary Mean \$	Risk Group Summary N	Risk Group Summary Total \$	Risk Group Summary Mean \$	Risk Group Summary N
GROUP-1	\$19,959	\$407	49	\$30,245	\$432	70	\$15,696	\$285	55
GROUP-2	\$87,538	\$5,471	16	\$180,545	\$7,523	24	\$19,126	\$2,732	7
GROUP-3	\$92,810	\$4,641	20	\$173,946	\$5,436	32	\$24,524	\$1,168	21
GROUP-4	\$49,846	\$7,121	7	\$69,632	\$9,947	7	\$2,587	\$431	6
GROUP-5	\$18,503	\$4,626	4	\$56,263	\$8,038	7			
GROUP-7	\$148,702	\$148,702	1						

Total Amount Paid - Disease Group Risk Stratification - Total Population



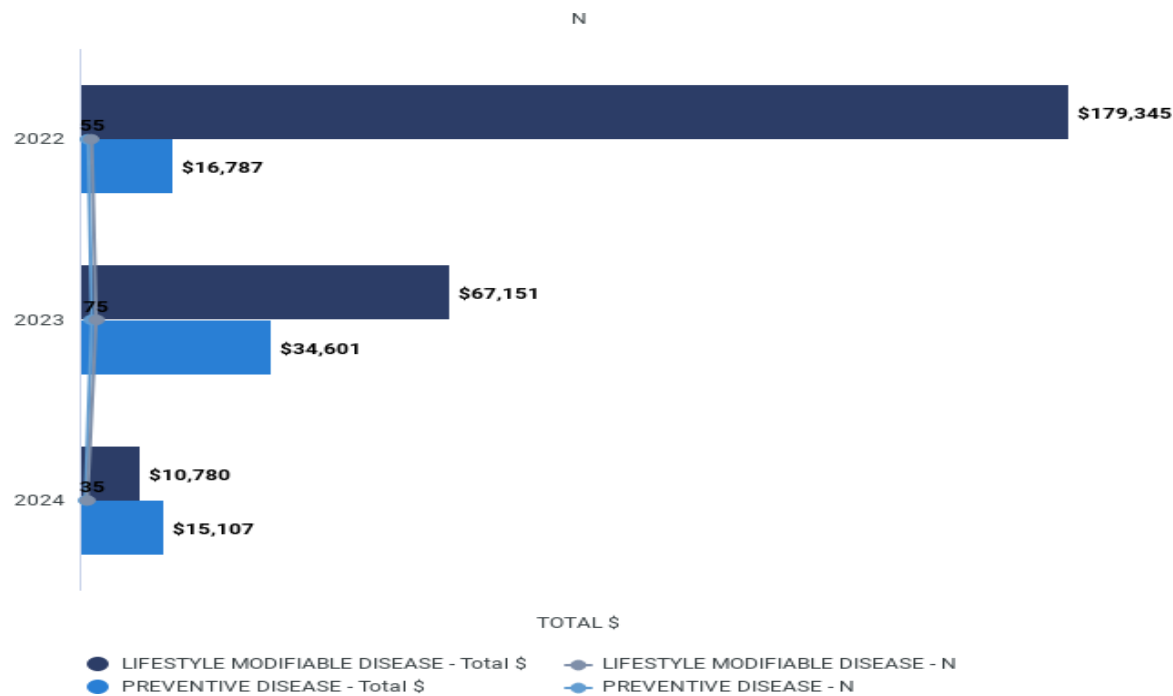
Mean Amount Paid - Disease Group Risk Stratification - Total Population



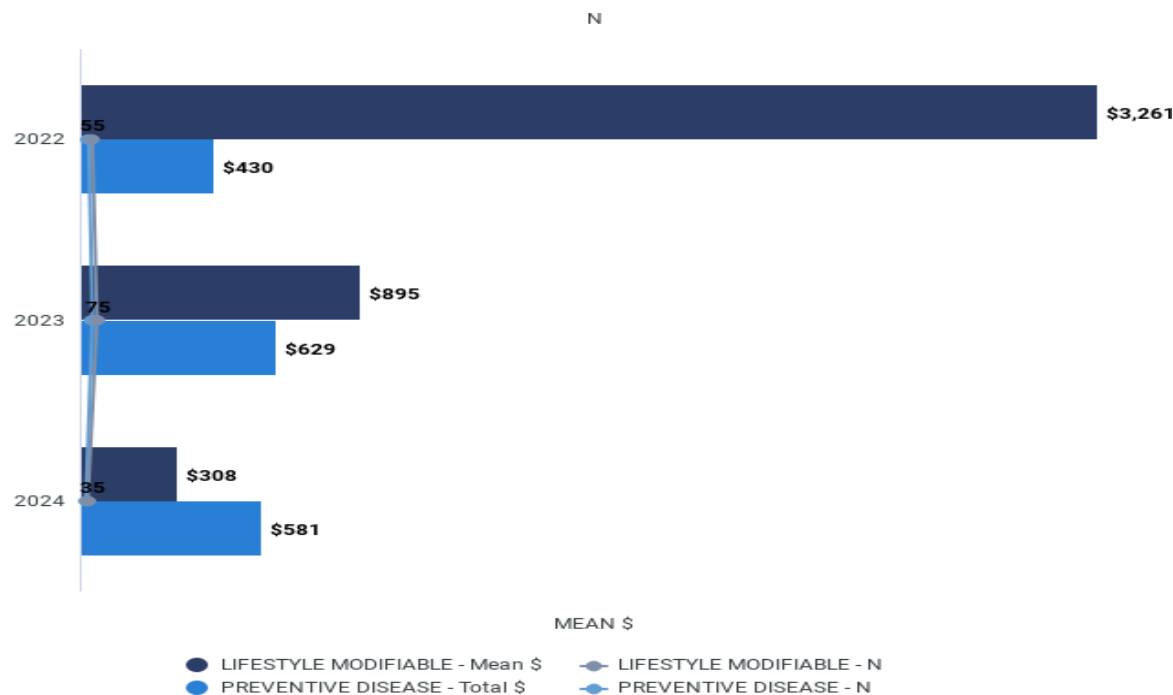
12. Expenditures Related to Lifestyle Modifiable & Preventive Utilization

Medical expenditures related to Lifestyle Modifiable and Preventive utilization were as follows:

Total Amount Paid - Lifestyle Modifiable and Preventive Utilization - Total Population



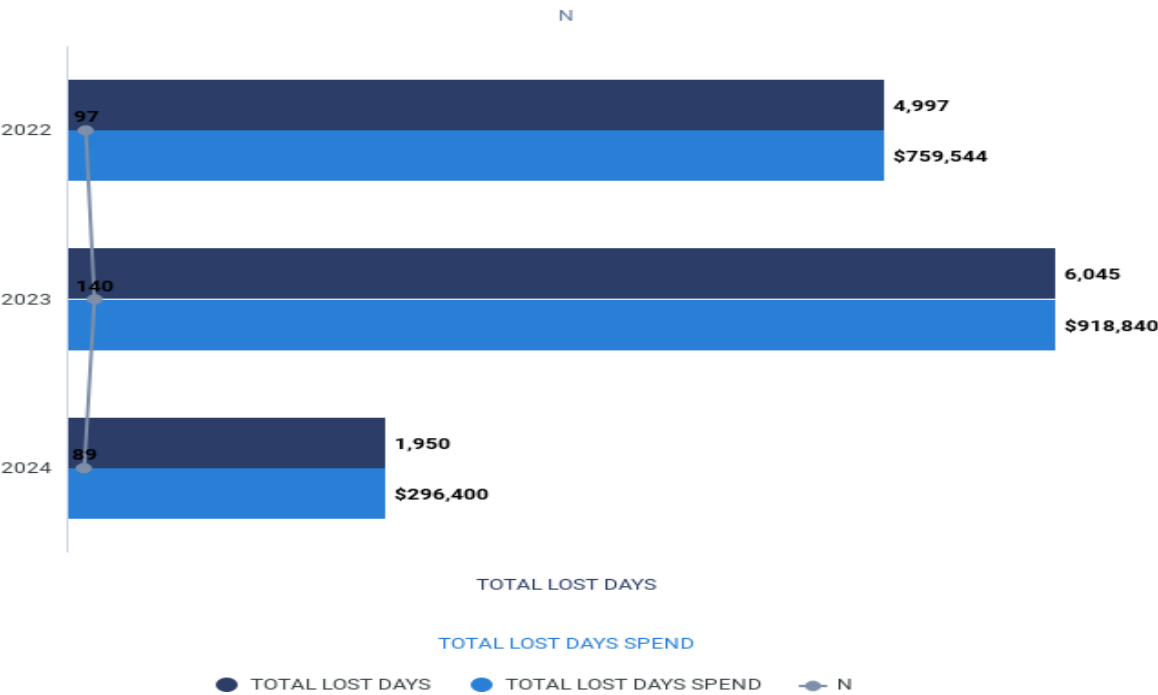
Mean Amount Paid - Lifestyle Modifiable and Preventive Utilization - Total Population



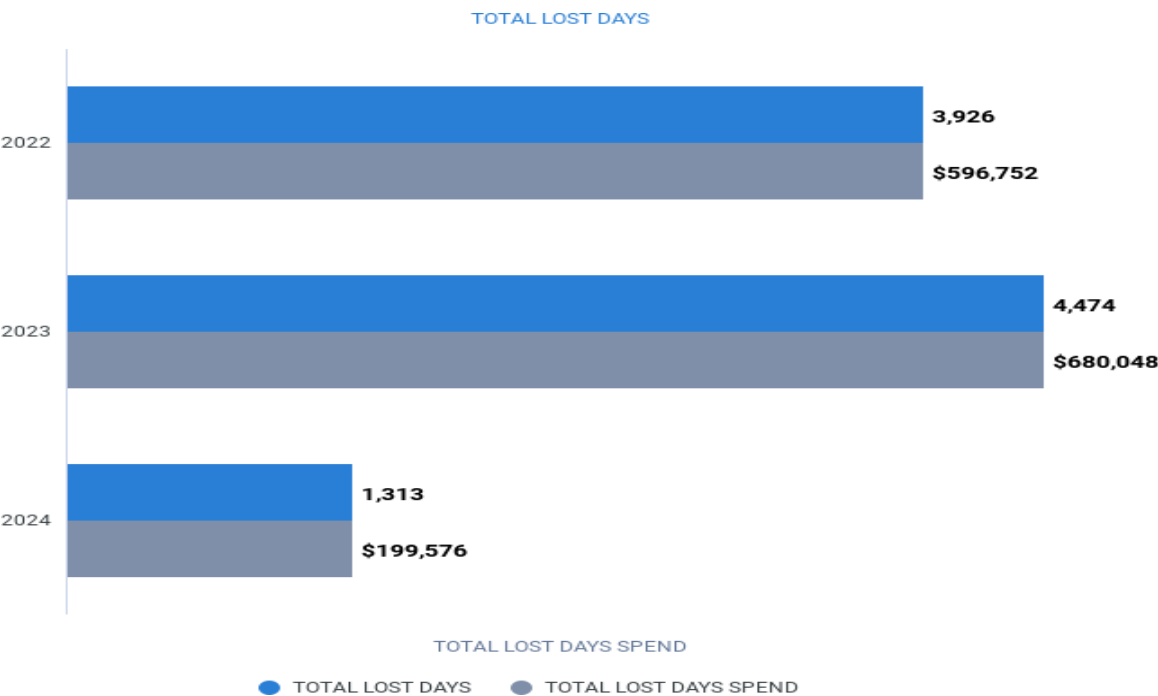
13. Estimated Lost Time & Cost due to Health Disparities

The calculation for lost time is based on normative data from the Work Loss Data Institute, which associates each ICD-9 and/or ICD-10 code (i.e., unique diagnosis) with a mean, or average, lost time. This analysis utilizes that data to estimate the total lost days related to health disparities. Total Amount Lost is calculated based on a rate of \$19 per hour for an 8-hour day.

Total Estimated Lost Days and Amount Lost due to Health Disparities - Total Population

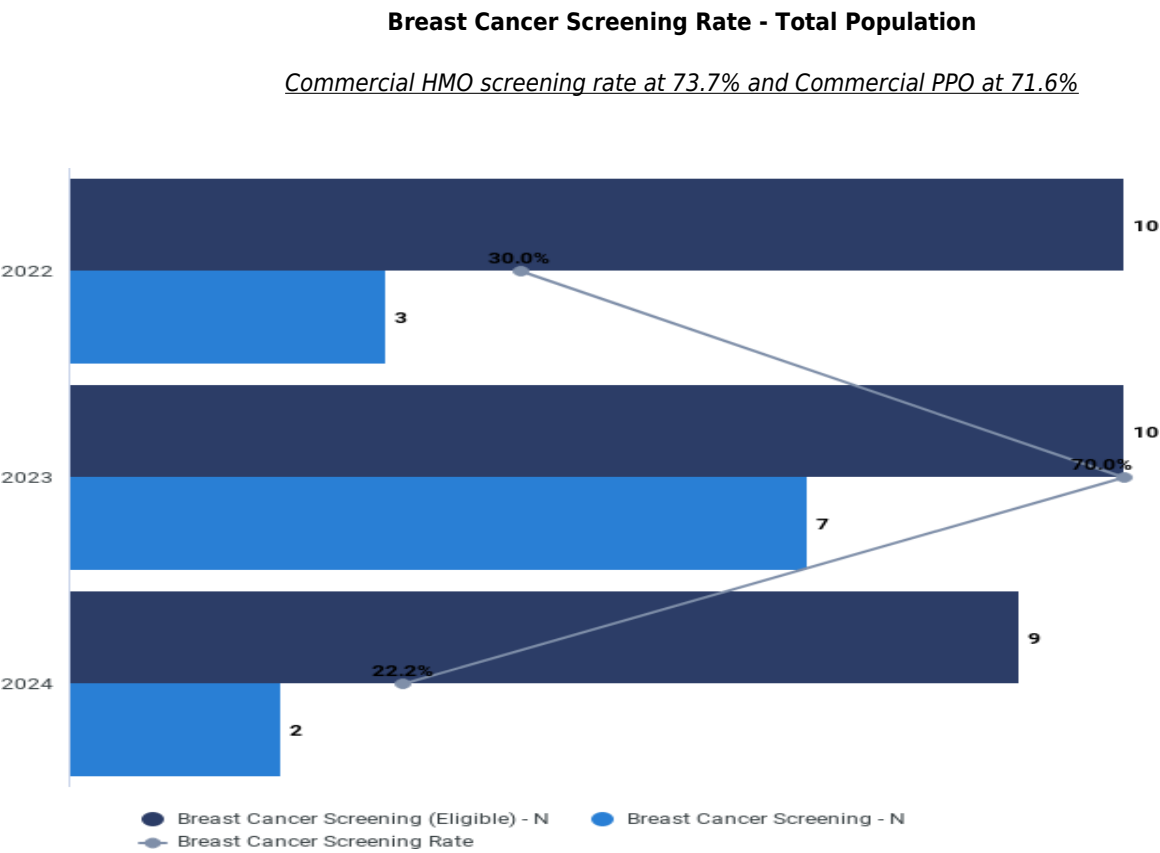


Total Estimated Lost Days and Amount Lost due to Health Disparities - Employees Only



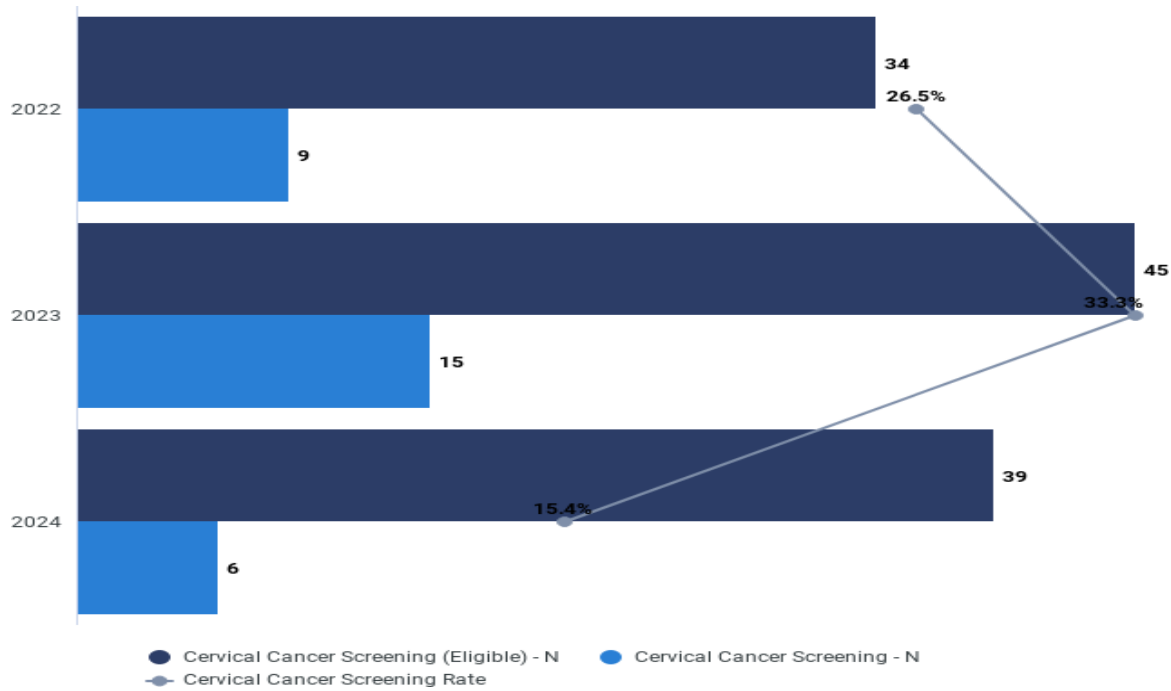
14. Preventive Screening Compliance

Preventive screening compliance* for breast cancer, cervical cancer, and colon cancer was well below HEDIS national guidelines. Screening rates were as follows:



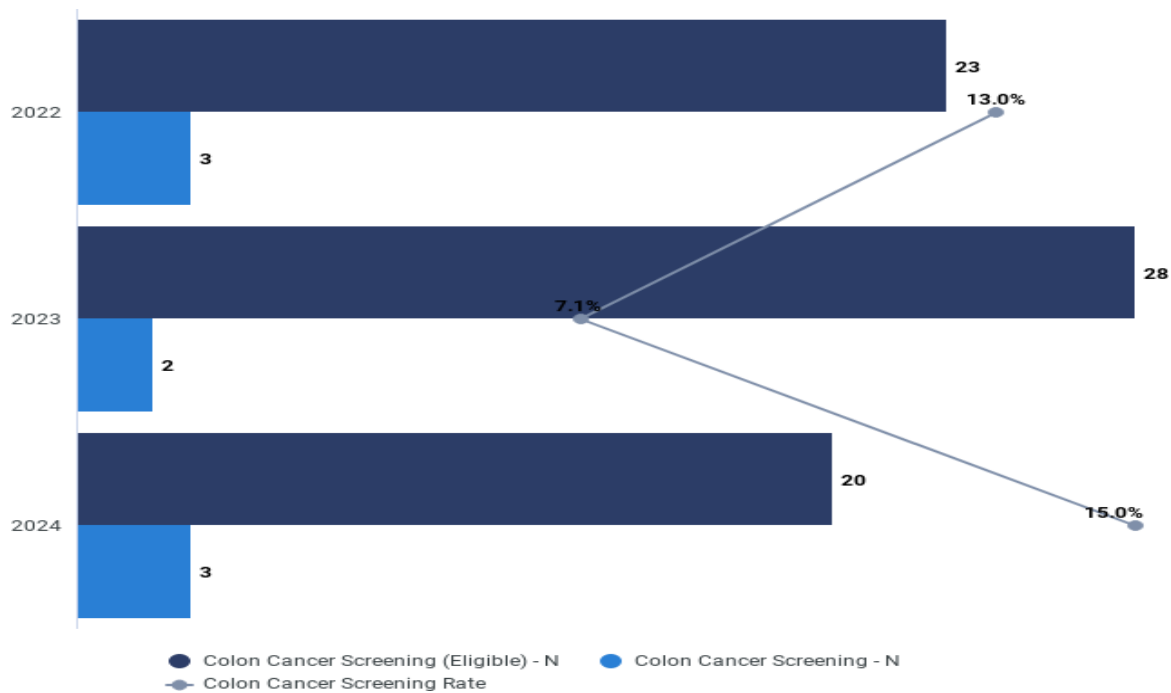
Cervical Cancer Screening Rate - Total Population

Commercial HMO screening rate at 76.2% and Commercial PPO at 74.2%



Colon Cancer Screening Rate - Total Population

Commercial HMO screening rate at 65.0% and Commercial PPO at 61.8%



15. Value of Preventive Screenings

The following section illustrates the value of preventive screenings within the population. The analyses identified preventive screenings that resulted in the biopsy or removal of suspicious polyps or tumors or resulted in a cancer diagnosis. Appendix 4 provides codes and definitions related to these analysis.

Number of Breast/Cervical/Colon cancer diagnosis identified after the screening*:

EBR Measures Year	EBR Measures Breast Cancer Screening - N	EBR Measures Cervical Cancer Screening - N	EBR Measures Colon Cancer Screening - N
2022	4	8	3
2023	7	15	0
2024	2	5	2

This analysis identified cancer diagnoses that presented in the claims data after the preventive screening occurred. It is plausible, although not certain, that the cancer diagnosis resulted from the preventive screening.

Breast Cancer Screenings that Resulted in Biopsy or Removal of Suspicious Polyps or Tumors*

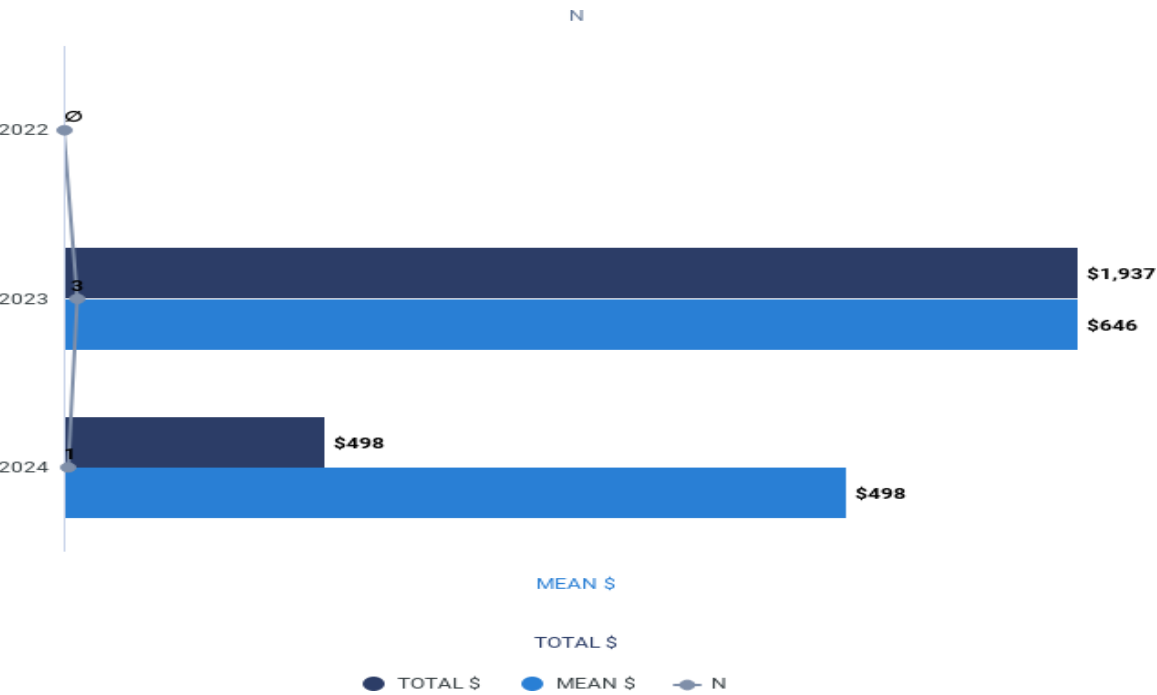
Medical records Reporting Year	Medical records TOTAL \$	Medical records MEAN \$	Medical records N
2022			
2023			
2024			
Total	\$0	\$0	0

Cervical Cancer Screenings that Resulted in Biopsy or Removal of Suspicious Polyps or Tumors*

Medical records Reporting Year	Medical records TOTAL \$	Medical records MEAN \$	Medical records N
2022			
2023			
2024			

Total	\$0	\$0	0
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Colon Cancer Screenings that Resulted in Biopsy or Removal of Suspicious Polyps or Tumors



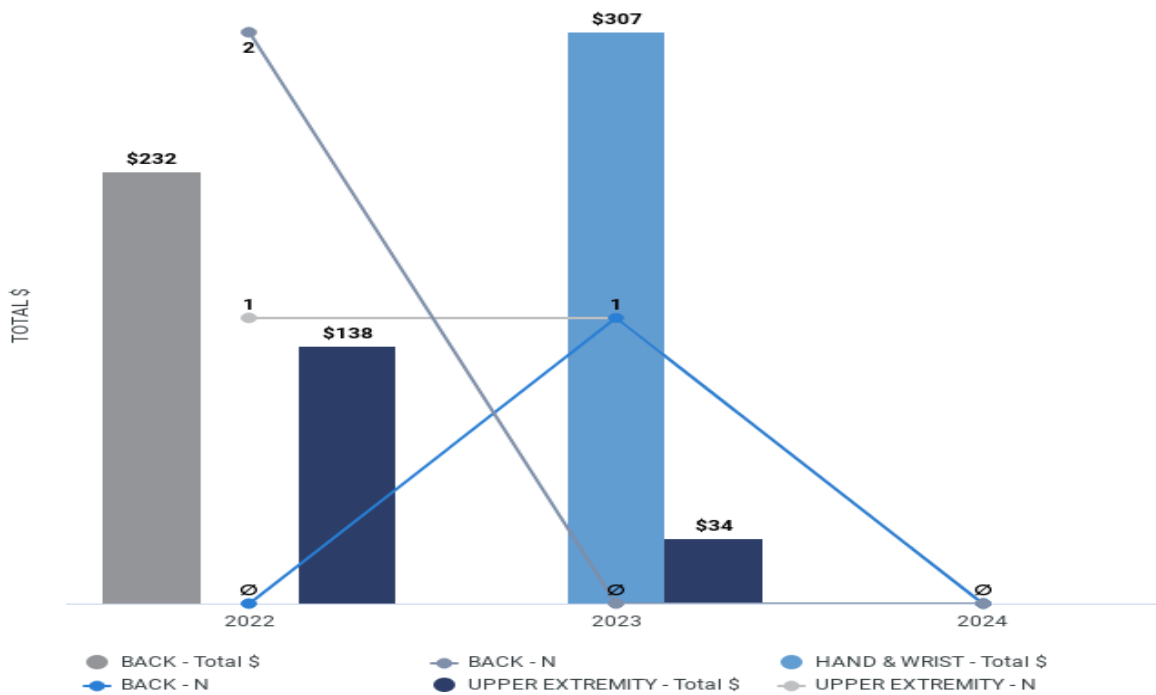
16. Potentially Work-Related Musculoskeletal Expenditures

The musculoskeletal conditions that may be caused by (non-accidental) physical work activities include disorders of inflammation, degeneration, and physiological disruption of muscles, tendons, ligaments, nerves, synovia, and cartilage involving limbs and trunk. Common examples are low back strain, tenosynovitis, and carpal tunnel syndrome.

Potentially work-related expenditures associated with the musculoskeletal diagnostic category were as follows:

Medical records Reporting Year	2022			2023			2024		
Medical records WORK RELATED MSK DIAGNOSIS	Medical records TOTAL \$	Medical records MEAN \$	Medical records N	Medical records TOTAL \$	Medical records MEAN \$	Medical records N	Medical records TOTAL \$	Medical records MEAN \$	Medical records N
HAND & WRIST				\$307	\$307	1			
BACK	\$232	\$116	2						
UPPER EXTREMITY	\$138	\$138	1	\$34	\$34	1			

Total Amount Paid - Potentially Work-Related Musculoskeletal Injuries - Employees Only



17. Catastrophic Claims

Medical expenditures related to catastrophic claims* were as follows:

**For this analysis, Catastrophic claims are defined as claims greater than \$100,000.*

Medical records Reporting Year	Medical records TOTAL \$	Medical records MEAN \$	Medical records N
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Total Amount Paid - CATASTROPHIC Claims - Total Population

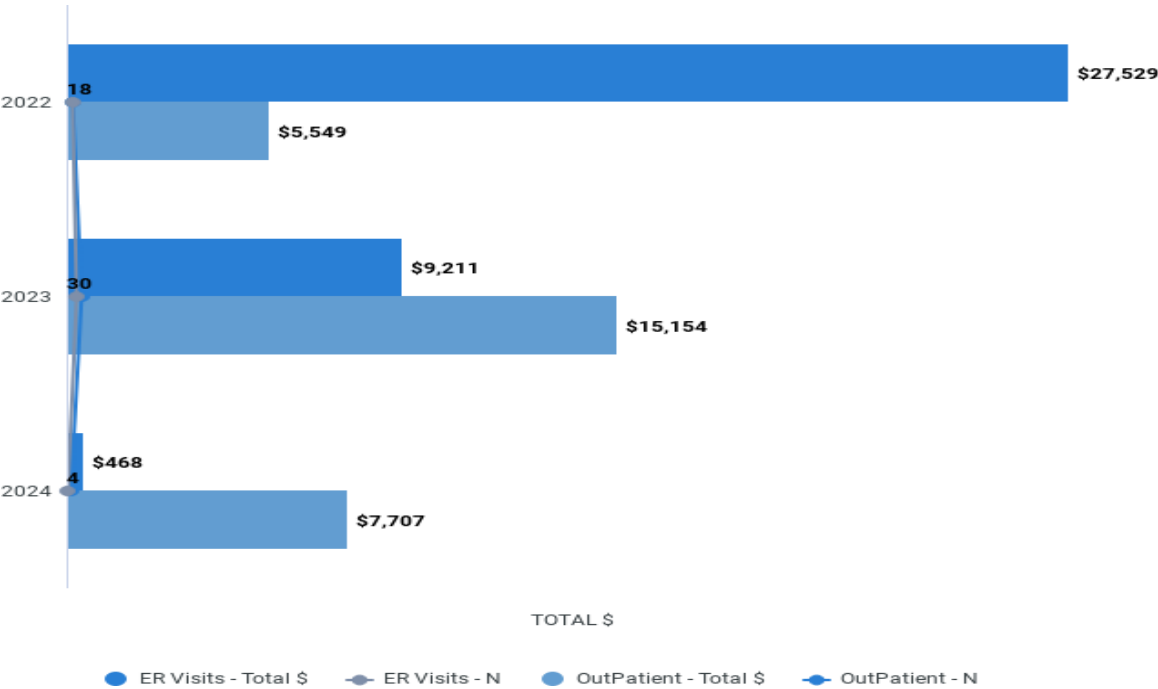
Medical records Reporting Year	Medical records TOTAL \$	Medical records N
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18. Inpatient, Outpatient & Emergency Room Expenditures

Medical expenditures related to Inpatient, Outpatient, and Emergency Room utilization were as follows:

Medical records InPatient/ OutPatient/ ER_Visits	ER Visits			OutPatient		
Medical records Reporting Year	Medical records TOTAL \$	Medical records MEAN \$	Medical records N	Medical records TOTAL \$	Medical records MEAN \$	Medical records N
2022	\$27,529	\$1,529	18	\$5,549	\$292	19
2023	\$9,211	\$307	30	\$15,154	\$337	45
2024	\$468	\$117	4	\$7,707	\$593	13

Total Amount Paid - Inpatient, Outpatient & Emergency Room Visits



19. Avoidable Emergency Room Visits

Examples of avoidable Emergency Room visits:

Medical records Reporting Year	2022	2023	2024	Total
Medical records DIAGNOSIS DESCRIPTION	Medical records TOTAL VISITS	Medical records TOTAL VISITS	Medical records TOTAL VISITS	Medical records TOTAL VISITS
URINARY TRACT INFECTION, SITE NOT SPECIFIED (N390)		2		2
OTHER SOMATOFORM DISORDERS (F458)	1			1
CHRONIC SINUSITIS, UNSPECIFIED (J329)		1		1
Total	1	3		4

Medical expenditures related to avoidable Emergency Room visits* were as follows:

Medical records Reporting Year	Medical records TOTAL \$	Medical records MEAN \$	Medical records N
2022	\$264	\$264	1
2023	\$630	\$210	3
2024			
Total	\$894	\$298	3

20. Primary Care Physician & Specialty Expenditures

Medical expenditures related to Primary Care Physician and Specialty utilization were as follows:

Total Amount Paid - Primary Care Physician & Specialty - Total Population

Medical records PCP & SPECIALTY SERVICES	OTHER SERVICES		PRIMARY CARE PHYSICIAN SERVICES	
Medical records Reporting Year	Medical records TOTAL \$	Medical records N	Medical records TOTAL \$	Medical records N
2022	\$243,757	89	\$173,602	83
2023	\$320,676	109	\$189,954	124
2024	\$42,127	56	\$19,806	82

Mean Amount Paid - Primary Care Physician & Specialty - Total Population

Medical records PCP & SPECIALTY SERVICES	OTHER SERVICES		PRIMARY CARE PHYSICIAN SERVICES	
Medical records Reporting Year	Medical records MEAN \$	Medical records N	Medical records MEAN \$	Medical records N
2022	\$2,739	89	\$2,092	83
2023	\$2,942	109	\$1,532	124
2024	\$752	56	\$242	82

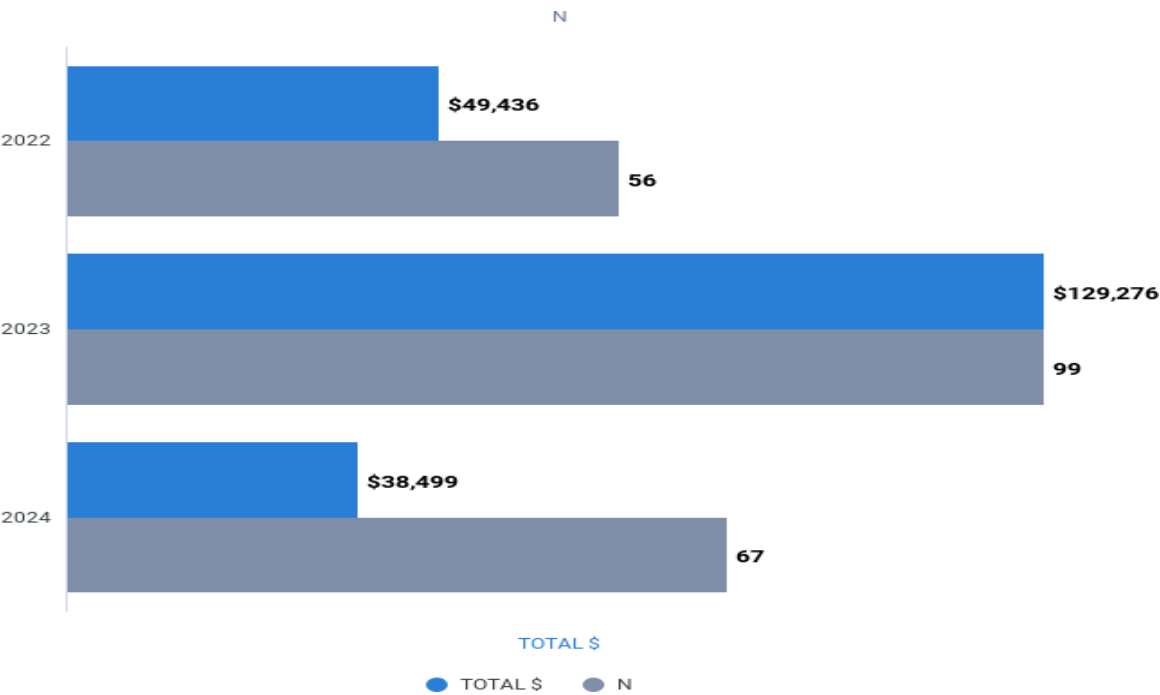
21. Overall Pharmacy Expenditures by Year

Pharmaceutical expenditures* (based on paid claims) were as follows:

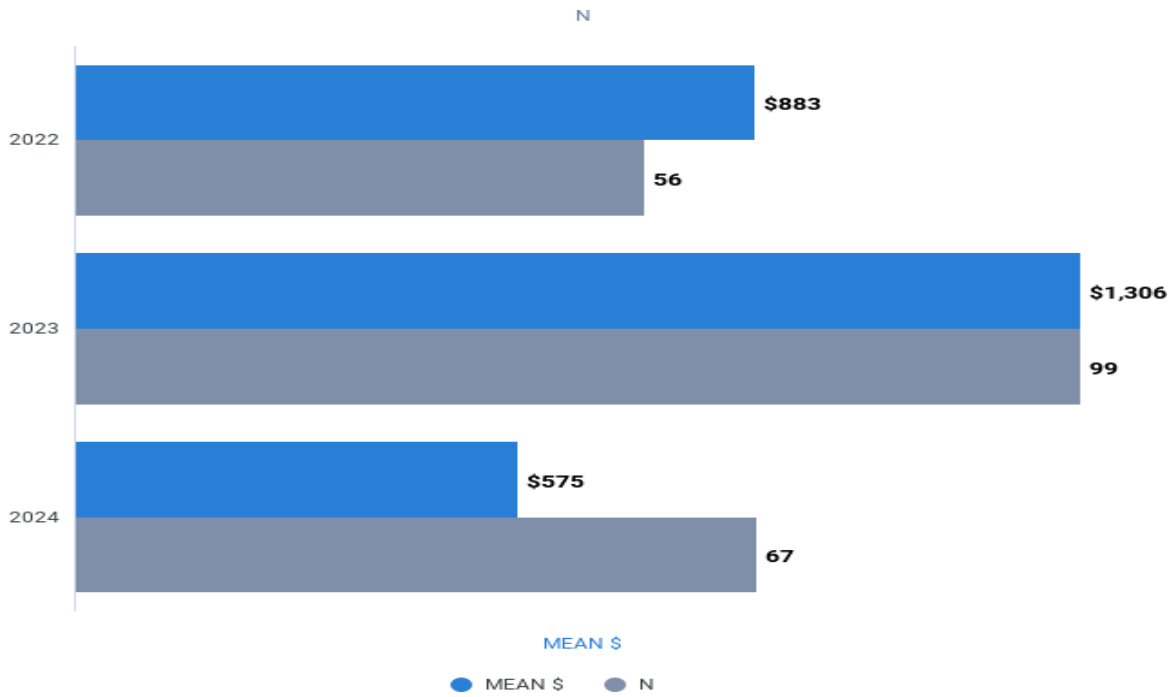
*Pharmacy expenditures do not include medical-related expenditures.

Pharmacy records Reporting Year	Pharmacy records TOTAL \$	Pharmacy records MEAN \$	Pharmacy records N
2022	\$49,436	\$883	56
2023	\$129,276	\$1,306	99
2024	\$38,499	\$575	67
Total	\$217,211	\$1,857	117

Total Amount Paid - Pharmacy - Total Population

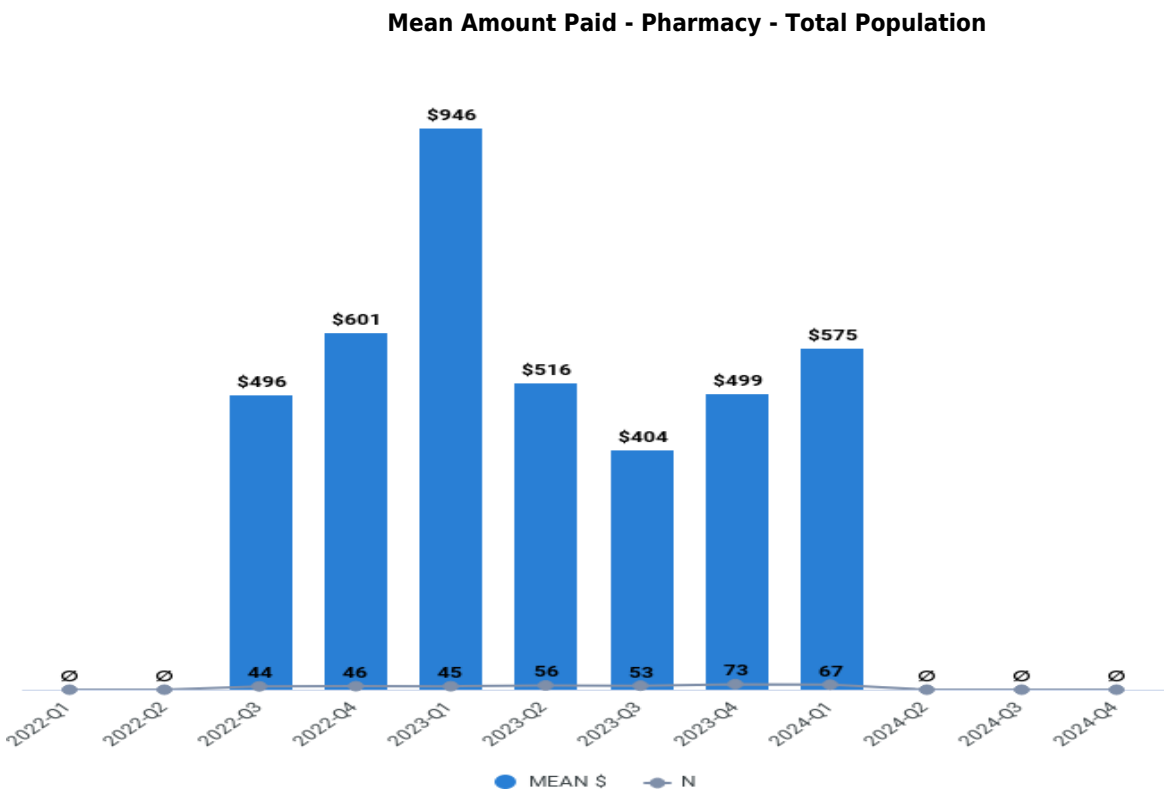
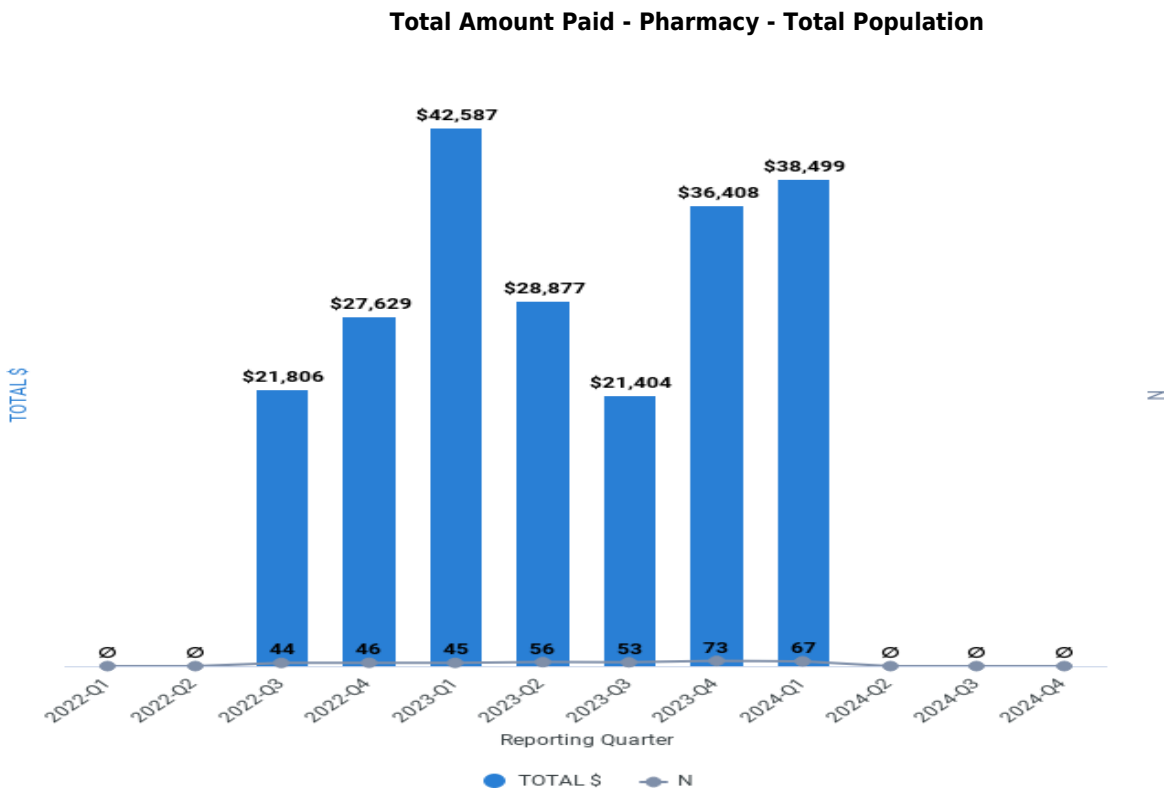


Mean Amount Paid - Pharmacy - Total Population



22. Overall Pharmacy Expenditures by Quarter

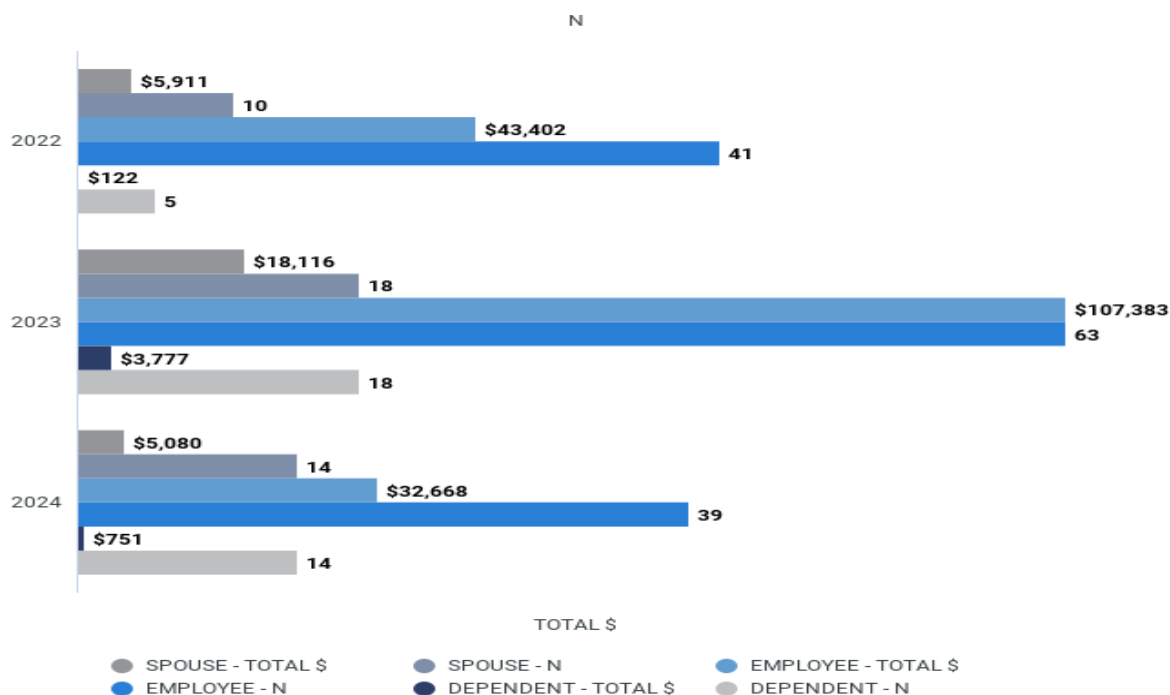
Pharmacy expenditures were as follows:



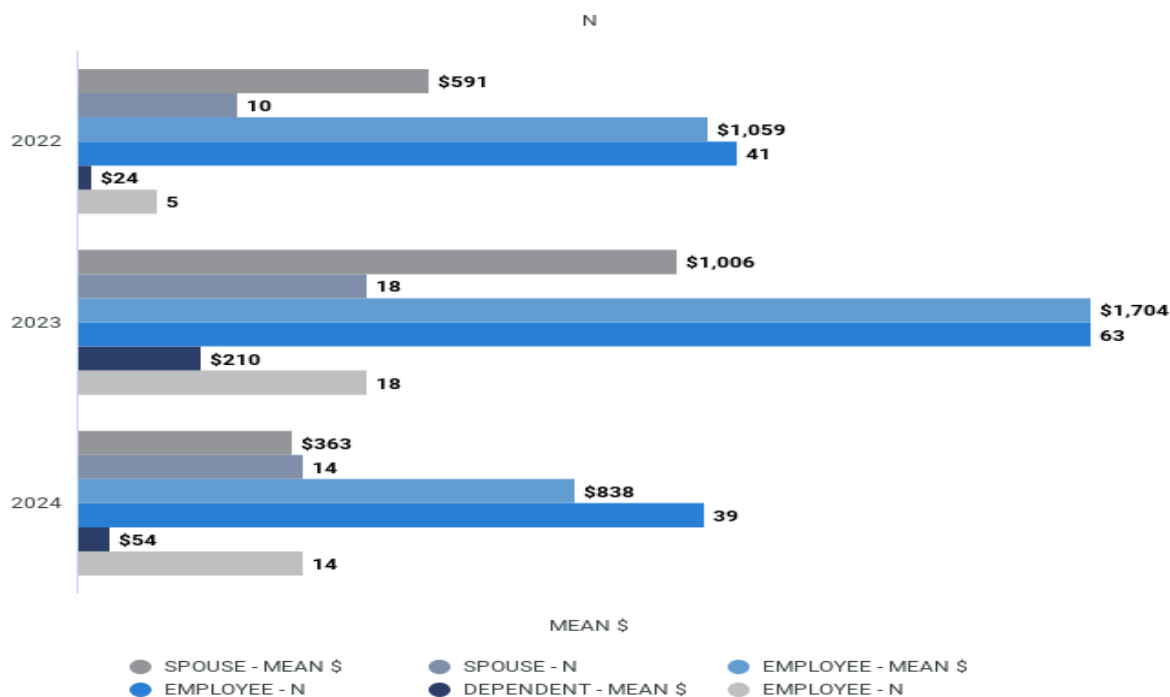
23. Employee/ Spouse/ Dependent Pharmacy Expenditures

Pharmaceutical expenditures related to Employees, Spouses, and Dependents were as follows:

Total Amount Paid - Pharmacy - Employee / Spouse / Dependent



Mean Amount Paid - Pharmacy - Employee / Spouse / Dependent



24. Medication Compliance

Medication Possession Ratio (MPR) is calculated as follows:

Days of Prescription = Beginning Date of Prescription – End Date of Prescription

Medication Possession Ratio = Sum of Days Supply for Prescription ÷ Days of Prescription

Medication Possession Ratio (MPR) measures the average compliance to prescriptions for those individuals who received a prescription and refilled it at least once.

Medication Possession Ratio for All Medication, Statin Medication, and Hypertension Medication was as follows:

Medication Possession Ratio for All Medication

Medication Possession Ratio Year	Medication Possession Ratio Medication Possession Ratio	Medication Possession Ratio Total Members
2022	94.5%	56
2023	92.1%	99
2024	98.9%	67
Total	94.2%	117

Medication Possession Ratio for Statin Medication

Medication Possession Ratio Year	Medication Possession Ratio Medication Possession Ratio	Medication Possession Ratio Total Members
2022	87.8%	13
2023	89.7%	17
2024	98.2%	12
Total	91.1%	18

Medication Possession Ratio for Hypertension Medication

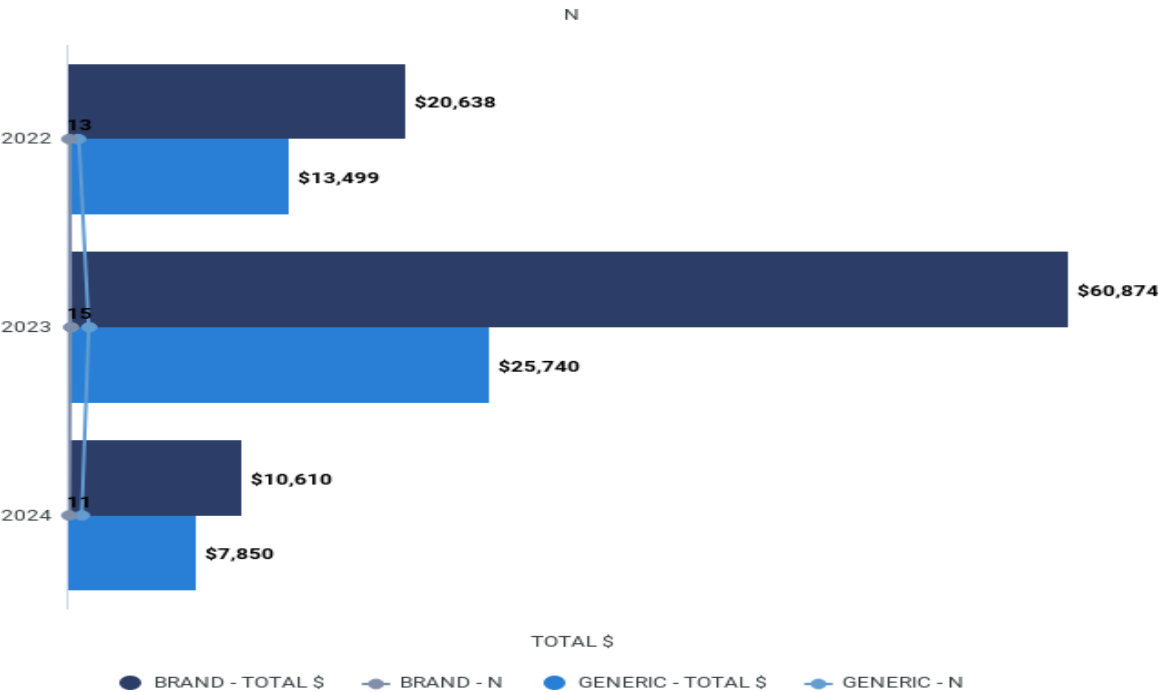
Medication Possession Ratio Year	Medication Possession Ratio Medication Possession Ratio	Medication Possession Ratio Total Members
2022	92.9%	13
2023	89.2%	26
2024	98.8%	19
Total	93.0%	33

25. Brand vs. Generic Medication Usage

Pharmacy expenditures related to Brand and Generic medication were as follows:

Pharmacy records Reporting Year	2022			2023			2024		
Pharmacy records BRAND/GENERIC	Pharmacy records TOTAL \$	Pharmacy records MEAN \$	Pharmacy records N	Pharmacy records TOTAL \$	Pharmacy records MEAN \$	Pharmacy records N	Pharmacy records TOTAL \$	Pharmacy records MEAN \$	Pharmacy records N
BRAND	\$20,638	\$1,588	13	\$60,874	\$4,058	15	\$10,610	\$965	11
GENERIC	\$13,499	\$275	49	\$25,740	\$292	88	\$7,850	\$131	60

Brand vs. Generic Medication Usage - Total Population



26. Appendix 1: Definition of Terms

The following section provides definitions of terms used throughout this report.

- **Acute Disease:** A disease with rapid onset and/or short duration.
- **Age/Gender Preventive Screenings:** Health screenings related to an individual's gender or age that have been found to help mitigate future risk of disease (e.g., screenings for breast cancer, cervical cancer, and colorectal cancer).
- **Chronic Disease:** A chronic condition is a health condition or disease that is persistent or otherwise long-lasting. The term chronic is usually applied when the disease lasts for more than three months. Common chronic diseases include Diabetes, Heart Disease, Hypertension, Asthma, COPD, and Cancer.
- **Co-Morbidity:** More than one disease or condition present in the same person at the same time (e.g., Diabetes with a co-morbidity of Hypertension). Conditions described as co-morbidities are often chronic or long-term conditions.
- **Confidence Interval:** Provides a range of values that is likely to include the actual value of an unknown population parameter. Associated with the interval is a measure of the confidence we have that the interval does indeed contain the parameter of interest.
- **Correlation Coefficient (p):** Measures the degree of linear association between two variables and is a number between -1 and 1. If there is perfect linear relationship with positive slope between two variables, they have a correlation coefficient of 1. If there is perfect linear relationship with negative slope between two variables, they have a correlation coefficient of -1. A correlation coefficient of 0 means that there is no linear relationship between two variables.
- **CPT:** The Current Procedural Terminology code set is a medical code set maintained by the American Medical Association. The CPT code set describes medical, surgical, and diagnostic services and is designed to communicate uniform information about medical services and procedures. CPT coding is similar to ICD coding, except that it identifies the services rendered rather than the diagnosis on the claim.
- **Disease-Specific Complication:** A secondary condition that develops during the course of a primary disease or condition (e.g., Ketoacidosis is a complication of Diabetes and Angina is a complication of Heart Disease).
- **Evidence-Based Medicine:** The conscientious, explicit, and judicious use of current best evidence in making decisions about the care of individual patients.
- **Gaps in Care:** Instances where an individual has failed to follow evidence-based guidelines for his or her health care. Examples of gaps in care include failure to comply with prescription medication usage and failure to have age/gender-specific preventive screenings.
- **Health Disparity:** The variation in rates of disease occurrence and disabilities between socioeconomic and/or geographically defined population groups.
- **ICD:** The International Statistical Classification of Diseases is a standard diagnostic coding tool for health classification. The World Health Organization maintains ICD. It provides a set of diagnostic codes for classifying diseases, including nuanced classifications of a wide variety of signs, symptoms, abnormal findings, complaints, social circumstances, and external causes of injury or disease. It is currently in its tenth revision (i.e., ICD-10).
- **Lifestyle Modifiable:** Lifestyle Modifiable illness includes forms of heart disease, cancer, diabetes, and various musculoskeletal conditions. Through a healthy lifestyle (e.g., exercise, weight management, nutrition, stress management, etc.), these diseases are preventable or their severity is modifiable through therapeutic lifestyle change. [The term is based on a multivariate analysis of 223,461 employee lives (i.e., 7 industries) conducted by the University of Michigan. Modifiable health risks were associated with approximately 26% of health care costs. High body mass index was the most prevalent lifestyle modifiable health risk. A total of 10 risks were included within the study.]

- **Mean:** The average of a set of numbers (i.e., the sum of the set of numbers divided by the total count of the numbers).
- **Median:** The midpoint (or middle) of a distribution of values in a data set. 50% of outcomes are on either side of this point. Provides a good measure of central tendency when there are outliers in a data set.
- **Mode:** The most frequently occurring value in a data set.
- **Morbidity:** The presence of a disease for an individual (e.g., Diabetes).
- **Multi-Morbidity:** The co-occurrence of multiple chronic or acute diseases and medical conditions within one person (e.g., Diabetes with Heart Disease).
- **N:** The size of a population or sample. (The N on each chart indicates the number of individuals incorporated within that particular analysis.)
- **Participants:** Individuals involved in a particular activity (e.g., individuals taking part in a wellness intervention).
- **Population:** A body of persons or individuals having a quality or characteristic in common. Also, a group of individual persons from which samples are taken for statistical measurement.
- **R-Square (R²):** Square of the correlation coefficient. The proportion of variability in one series that can be explained by the variability of one or more other series in a regression model. It is the measure of quality of fit, with 100% meaning perfect predictability. An R-Square of less than .05 is considered to be statistically significant.
- **Recurrent Disease:** Recurrent diseases relapse repeatedly, with periods of remission in between.
- **Regression Analysis:** Models relationships between variables and determines the magnitudes of those relationships. The models can be used to make predictions.
- **Regression toward the Mean:** The tendency of extreme values to move toward the average.
- **Risk:** The chance that an investment will lose value. “At-risk” is defined as a state marked by a high level of risk or susceptibility (e.g., patients at risk for a disease state).
- **Sample:** A collection of units from a population. In statistical studies, you analyze the sample to make inferences or draw conclusions about a target population.
- **Standard Deviation:** The square root of the variance. The most commonly used measure of spread as it converts the variance back into the same units in which the data is measured. If a data set contains values that are all very similar, the standard deviation will be small; if the data set is very spread out, the standard deviation will be large, indicating a greater variability or level of risk.
- **Variance:** The averaged squared distance or deviation of each data point from the mean.
- **Subscribers:** Individuals who have contracted for services under a health plan. (Not all subscribers on the health plan will have medical claims; only those individuals who utilized health care during the dates included in the analysis will be included in data related to claims expenditures.)

27. Appendix 2: Examples of Diagnostic Categories

The following section provides examples of the major diagnostic categories.

- **Certain Conditions Originating in the Perinatal Period** - Fetal Alcohol Syndrome, Slow Fetal Growth and Fetal Malnutrition, Failure to Thrive in Newborn
- **Complications of Pregnancy, Childbirth, and the Puerperium** - Ectopic Pregnancy, Pre-Eclampsia, Obstructed Labor
- **Congenital Anomalies** - Spina Bifida, Cleft Palate, Birthmarks, Down Syndrome
- **Diseases of the Blood and Blood-Forming Organs** - Sickle-Cell Anemia, Hemophilia, Elevated White Blood Cell Count
- **Diseases of the Circulatory System** - Rheumatic Fever, Hypertension, Myocardial Infarction, Pulmonary Heart Disease
- **Diseases of the Digestive System** - Gingivitis, Gastro esophageal Reflux, Appendicitis, Hernia, Necrosis of Liver, Pancreatitis
- **Diseases of the Genitourinary System** - Renal Failure, Urinary Tract Infection, Inflammatory Diseases of Prostate, Mastitis
- **Diseases of the Musculoskeletal and Connective Tissue** - Rheumatoid Arthritis, Sciatica, Tendonitis, Osteoporosis, Scoliosis
- **Diseases of the Nervous System and Sense Organs** - Bacterial Meningitis, Sleep Disorders, Alzheimer's, Parkinson's Disease, Chronic Pain, Multiple Sclerosis, Epilepsy, Retinal Detachment, Glaucoma, Strabismus
- **Diseases of the Respiratory System** - Sinusitis, Laryngitis, Pneumonia, Bronchitis
- **Diseases of the Skin and Subcutaneous Tissue** - Cellulitis, Eczema, Dermatitis
- **E Codes (External Causes of Injury)** - Motor Vehicle Traffic Accident Involving Collision with Other Vehicle, Dog Bite, Accident Caused by Firearm
- **Endocrine, Nutritional & Metabolic Diseases, and Immunity Disorders** - Disorders of the Thyroid Gland, Diabetes Mellitus, Hyperlipidemia
- **Infectious and Parasitic Diseases** - Cholera, Tuberculosis, HIV, Syphilis
- **Injury and Poisoning** - Dislocation of Shoulder, Sprains and Strains of Wrist and Hand, Concussion, Open Wound of Finger(s), Black Eye, Burns, Poisoning, Toxic Effect of Alcohol, Frostbite
- **Mental Disorders** - Dementia, Drug Withdrawal, Schizophrenic Disorders, Depression
- **Neoplasms** - Oral Cancer, Lung Cancer, Skin Cancer
- **Symptoms, Signs, and Ill-Defined Conditions** - Coma, Dizziness/Vertigo, Chronic Fatigue Syndrome, Rash (Unspecified), Headache, Sore Throat, Heart Palpitations, Cough, Nausea and Vomiting, Abdominal Pain
- **V Codes (Supplementary Classification of Factors Influencing Health Status and Contact with Health Services)** - Persons with Potential Health Hazards Related to Personal and Family History, Live-Born Infants, Persons Encountering Health Services for Specific Procedures or Aftercare, Persons Encountering Health Services in Other Circumstances, Body Mass Index

28. Appendix 3: Examples of Complications of Diabetes

The following section provides examples of complications associated with diabetes.

- **Cardiovascular** - Atherosclerosis, Hyperlipidemia, Fibrosis and Hypertrophy of Myocardium, Coronary Artery Disease
- **Cerebrovascular** - Stroke (Transient Ischemic Attack)
- **Nephropathy** - Damage to or disease of a kidney; Nephrosis is non-inflammatory nephropathy. Nephritis is inflammatory kidney disease.
- **Neuropathy** - Autonomic Neuropathies (ie., affects nerve function in bladder, bowel, heart, and sexual function) and Peripheral Neuropathies (i.e., affects nerves in hands, feet, legs, arms, and toes)
- **Peripheral Vascular Disease** - Commonly referred to as Peripheral Artery Disease (PAD) or Peripheral Artery Occlusive Disease (PAOD) or Peripheral Obliterative Arteriopathy; refers to the obstruction of large arteries not within the coronary, aortic arch vasculature, or brain. PVD can result from atherosclerosis, inflammatory processes leading to stenosis, an embolism, or thrombus formation. It causes either acute or chronic ischemia (i.e., lack of blood supply). Often PVD is a term used to refer to atherosclerotic blockages found in the lower extremity.
- **Retinopathy** - Due to persistent or acute damage to the retina of the eye. Ongoing inflammation and vascular remodeling may occur over periods of time where the patient is not fully aware of the extent of the disease. Frequently, retinopathy is an ocular manifestation of systemic disease as seen in diabetes or hypertension.

29. Appendix 4: Preventive Screening Eligibility Definitions

The following section lists the eligibility requirements used in Section: "Preventive Screening Compliance" of this report to identify individuals who completed preventive screenings for breast cancer, cervical cancer, and colon cancer.

Type of Screening & Eligibility for Screening* are:

Breast Cancer Screening: Females ages 40 to 74

Cervical Cancer Screening: Females ages 21 to 64

Colon Cancer Screening:** Males and females ages 50 to 75

**For this analysis, compliance was based on completion of the screening within a one-year period.*

***Includes Fecal Occult Blood Test, Flexible Colonoscopy, and Colonoscopy.*